

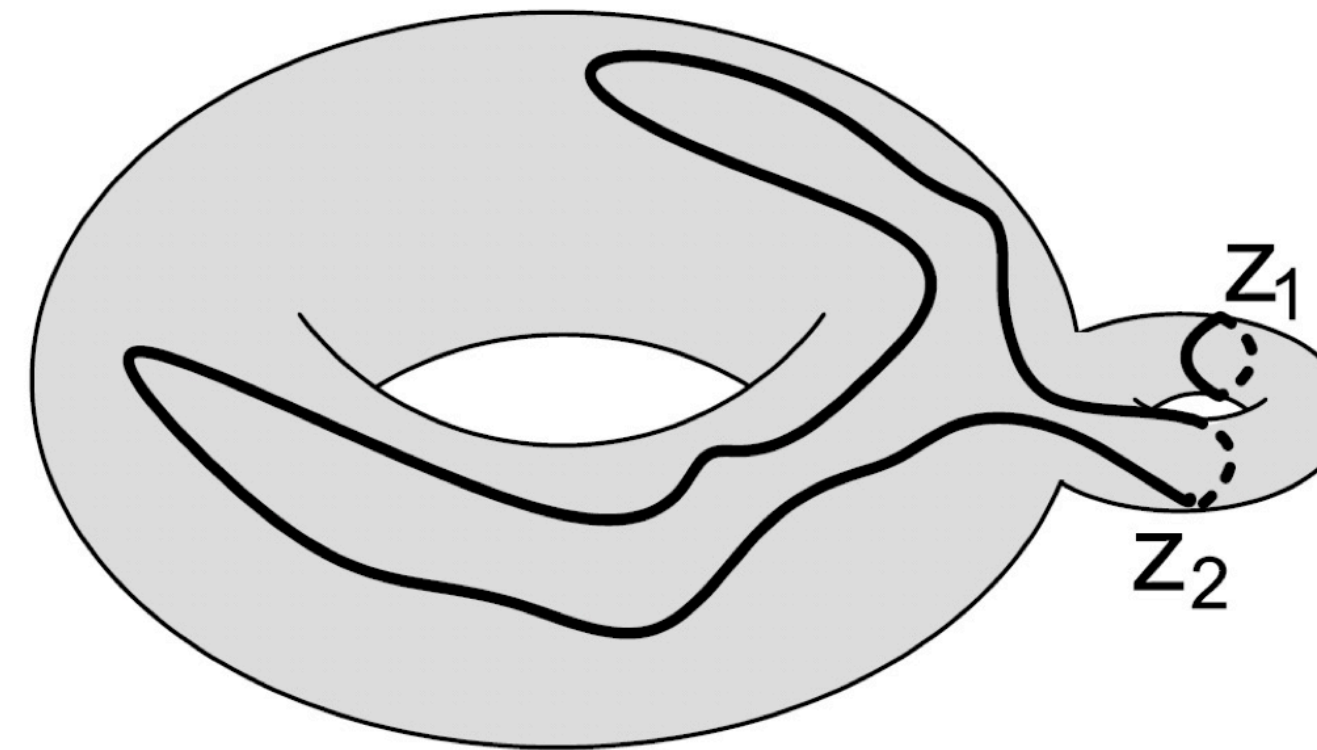
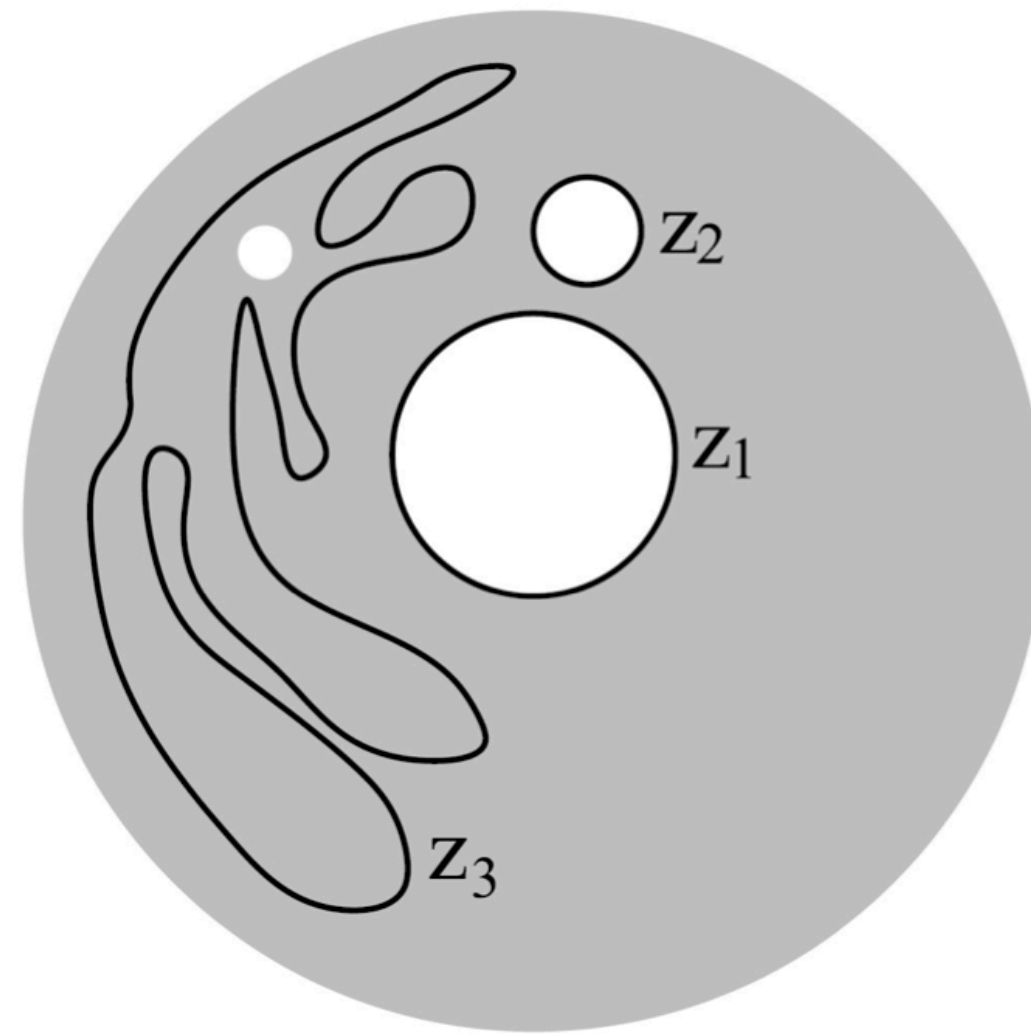
Topological Data Analysis

Lecture 11

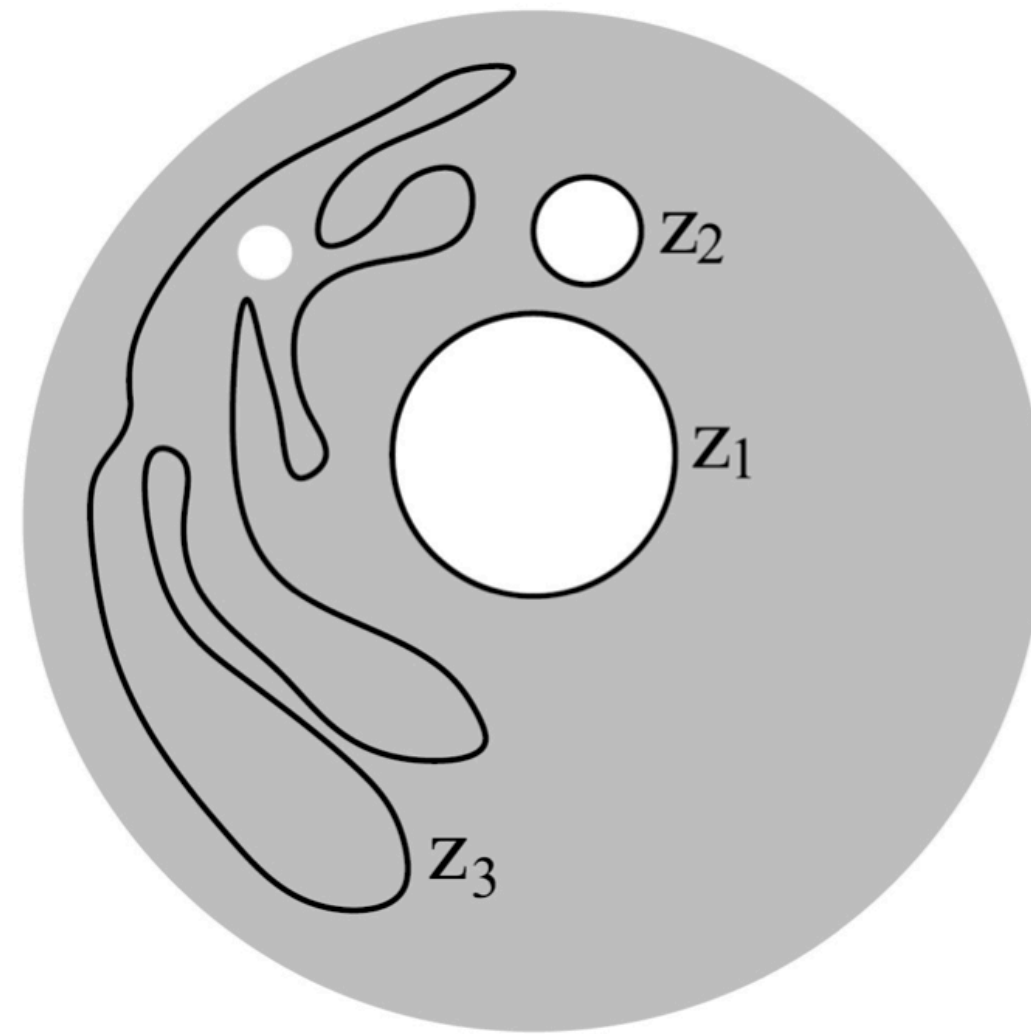
Homology representatives

Oleg Kachan

Homology representatives



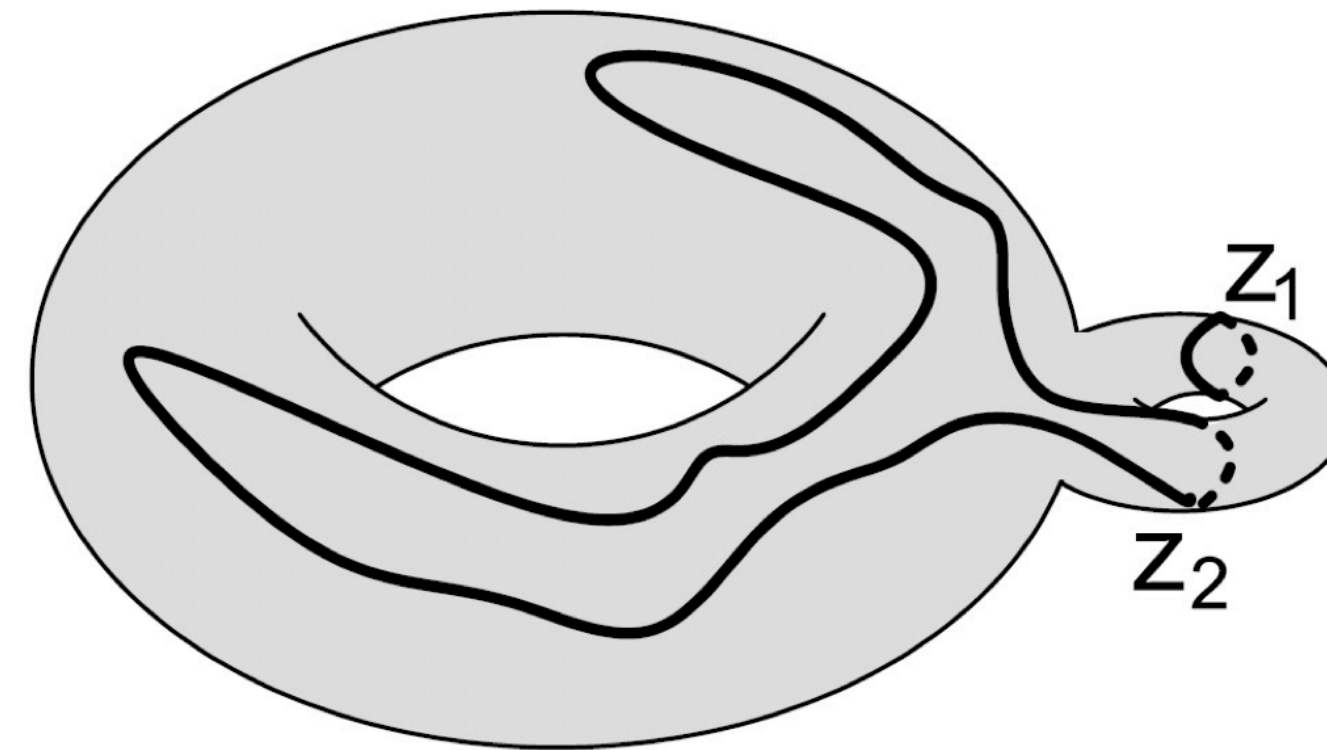
Homology representatives



$$z_1 \simeq z_2$$

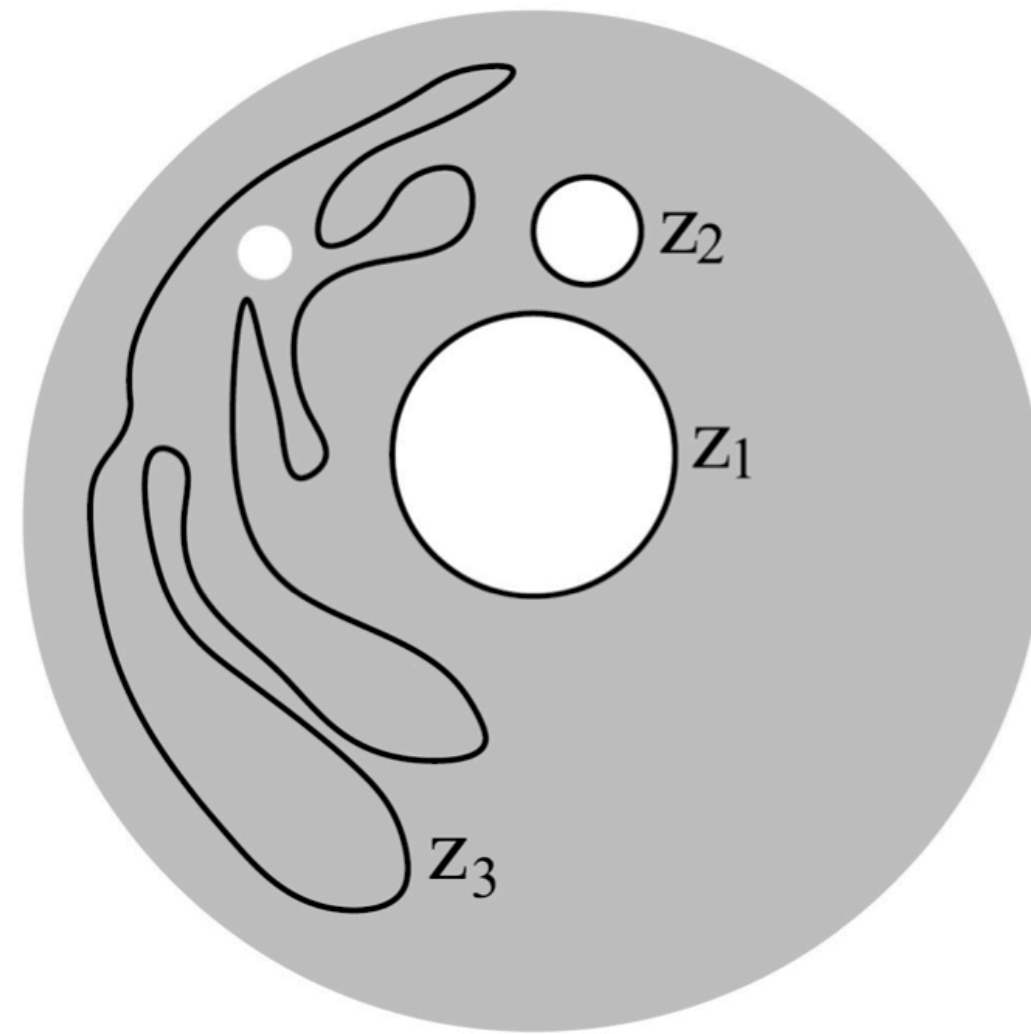
$$z_1 \simeq z_2$$

$$z_2 \simeq z_3$$



$$z_1 \sim z_2$$

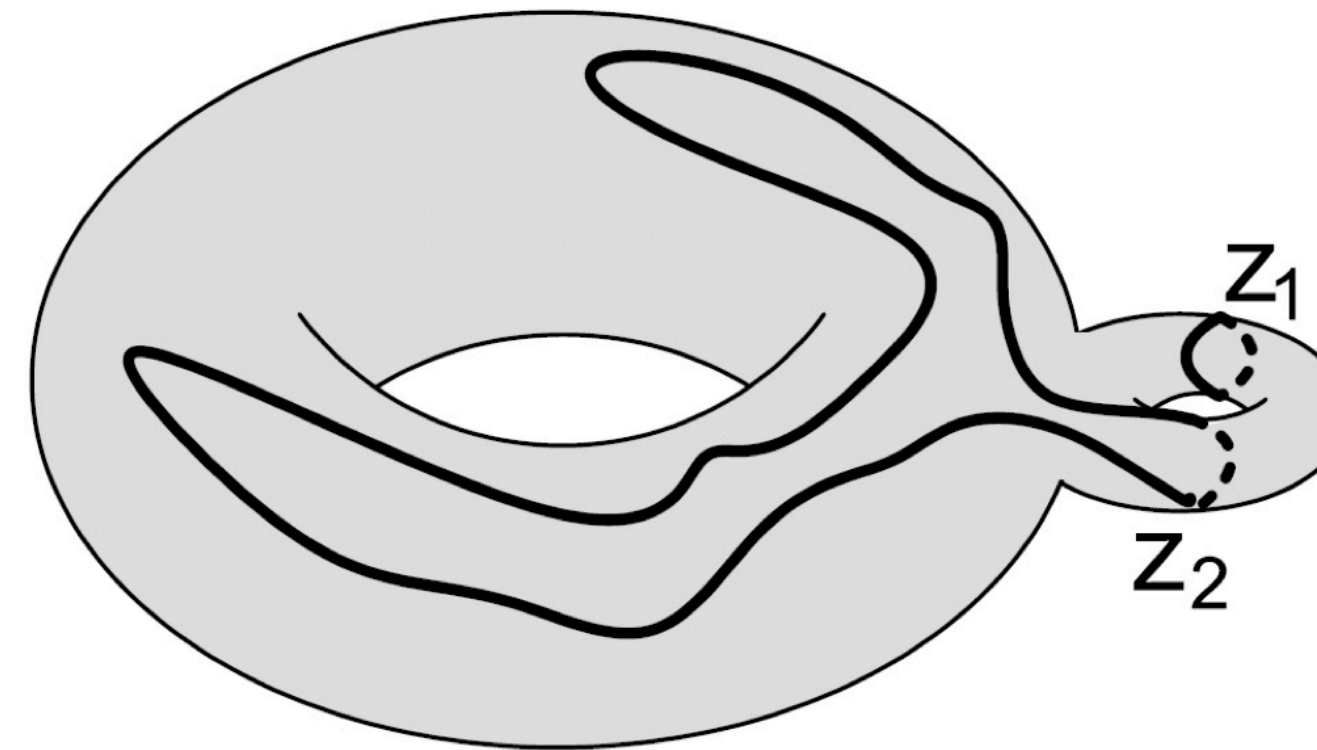
Homology representatives



$$z_1 \approx z_2$$

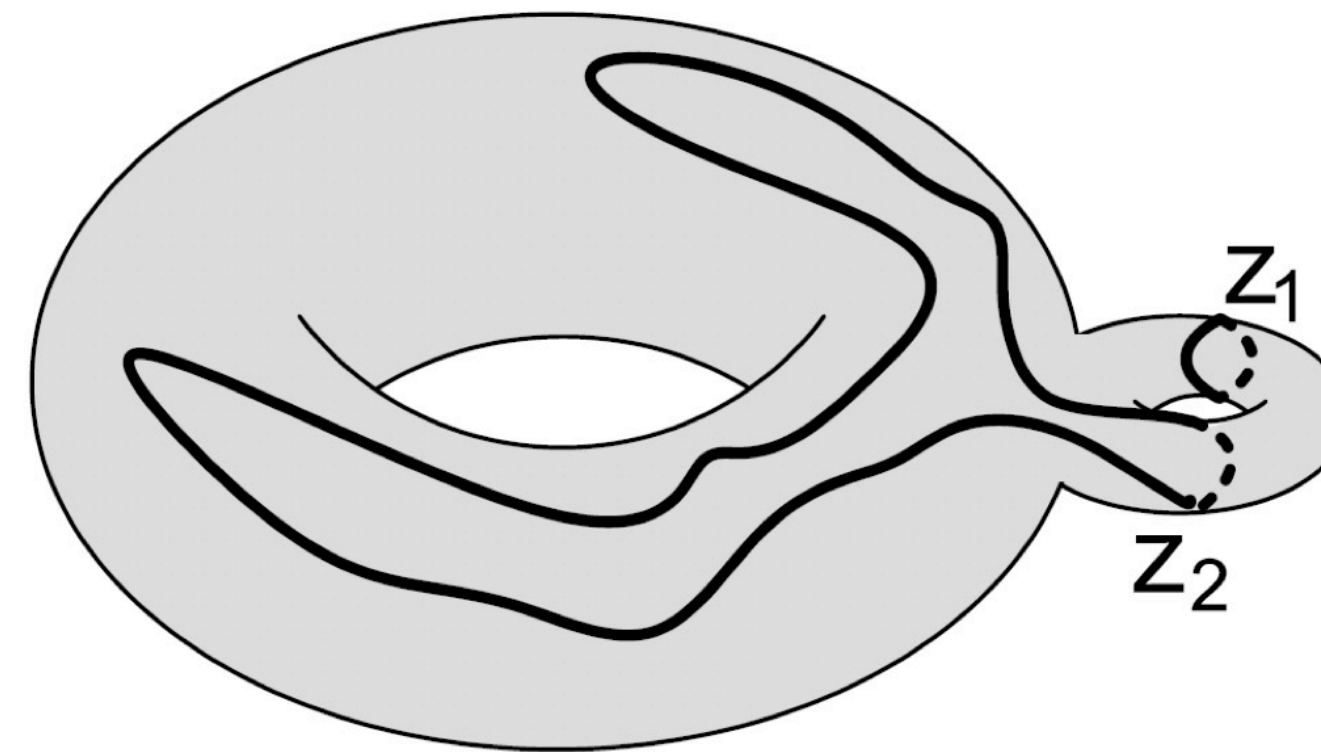
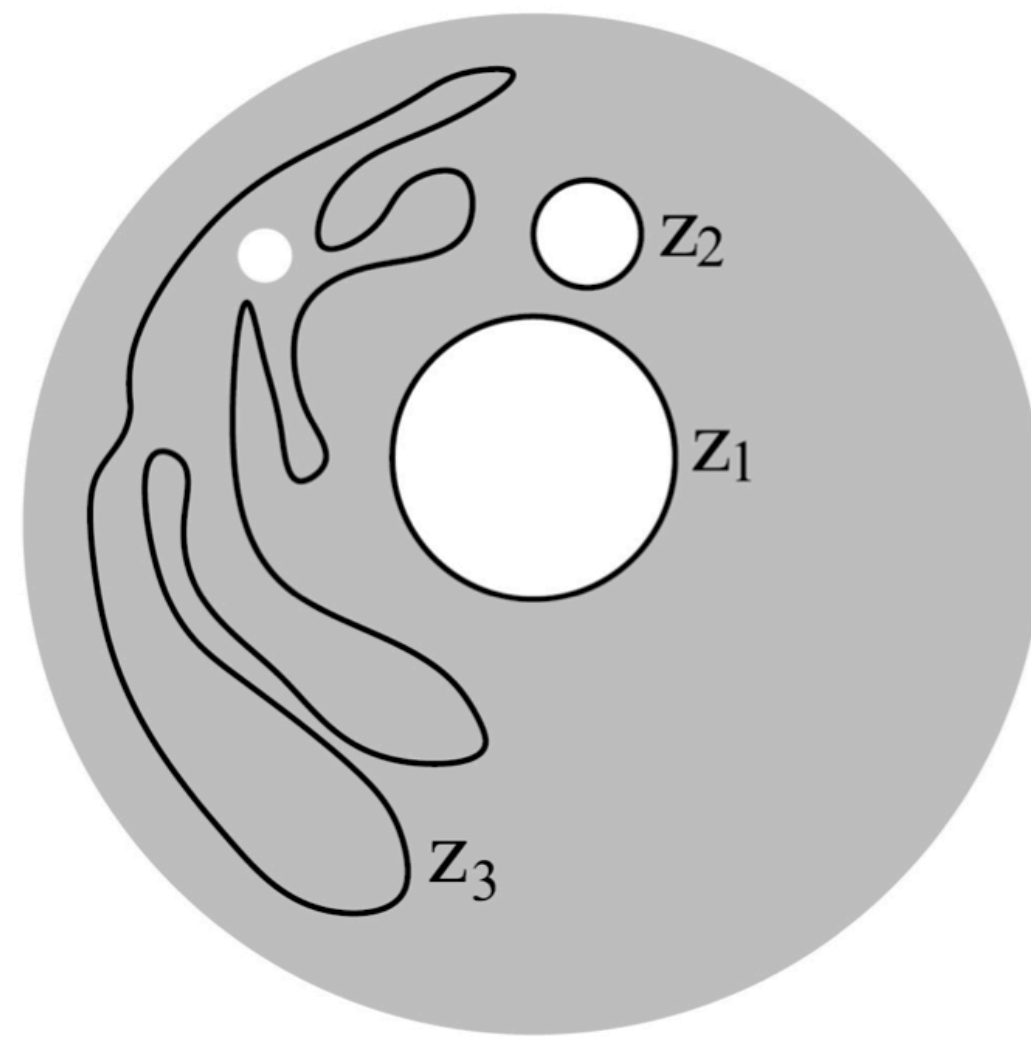
$$z_1 \approx z_2 \quad * \quad z_1, z_2 ; z_3$$

$$z_2 \approx z_3$$



$$z_1 \sim z_2 \quad * \quad z_1 ; z_2$$

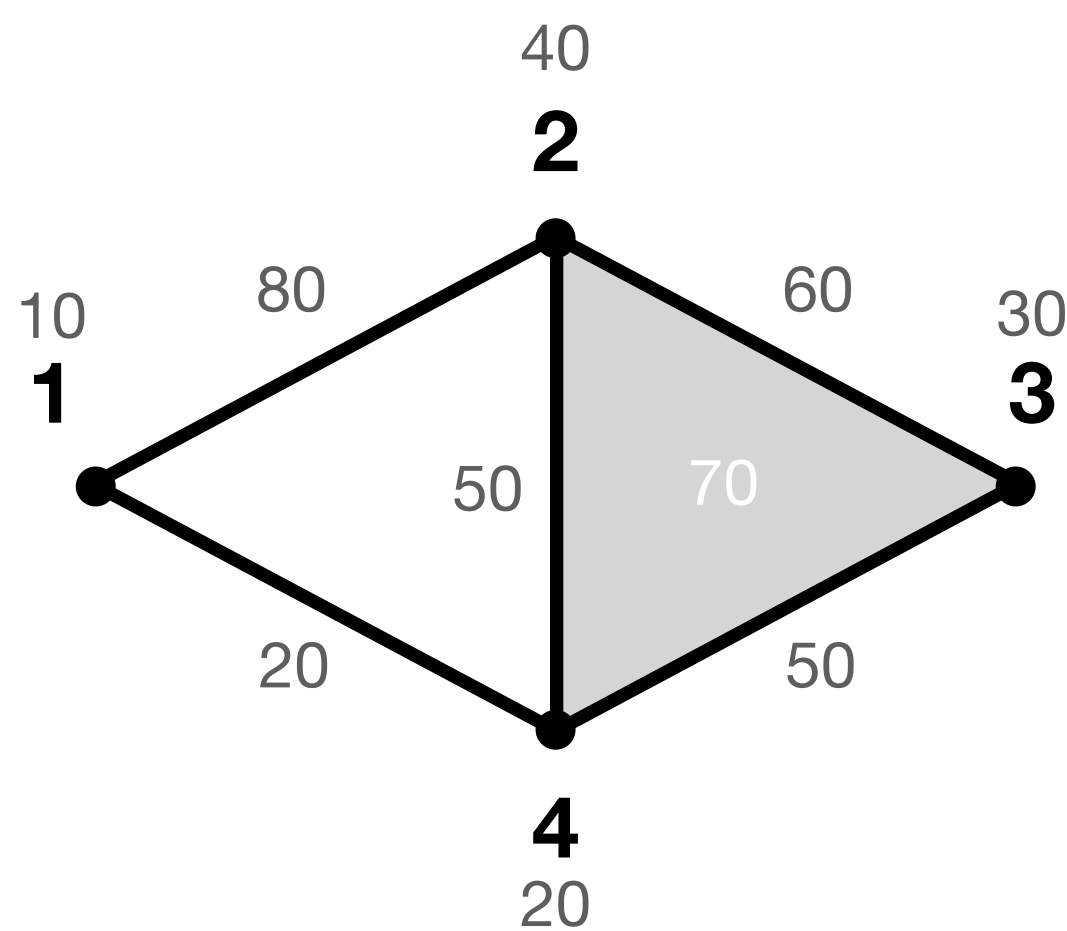
Homology representatives



Optimal cycle

$$z^* = \arg \min_z \ell(z_0) \quad s.t. \quad z \sim z_0$$

Boundary matrix reduction



B =

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
0										
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

Filtration function provides order on simplices, therefore on columns and rows of the filtration matrix. Ties are broken, first by simplex dimension, second by lexicographic order given by order on vertices.

Boundary matrix reduction

10
1 ●

	[1]									
	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=10

Boundary matrix reduction

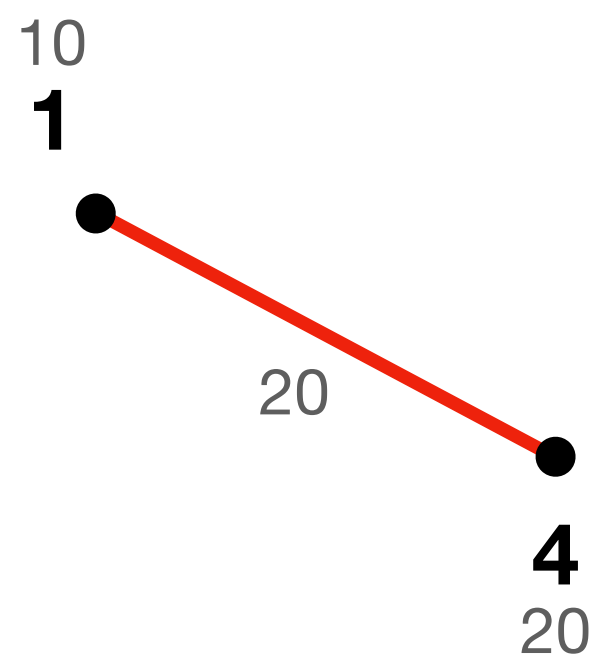
10
1
●

●
4
20

		[1] 10	[4] 20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12	
1			1								1
4			1			1	1				
14											
3							1	1			
2						1		1			1
24									1		
34									1		
23									1		
12											

t=20

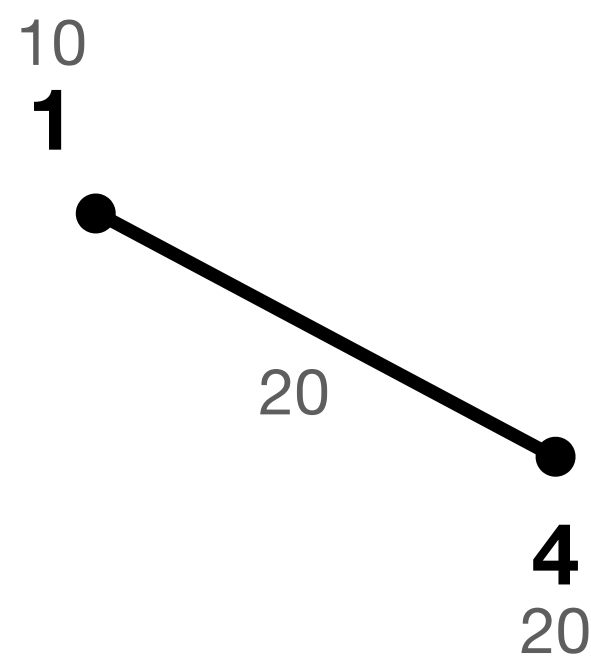
Boundary matrix reduction



	[1]		[4]							
	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=20

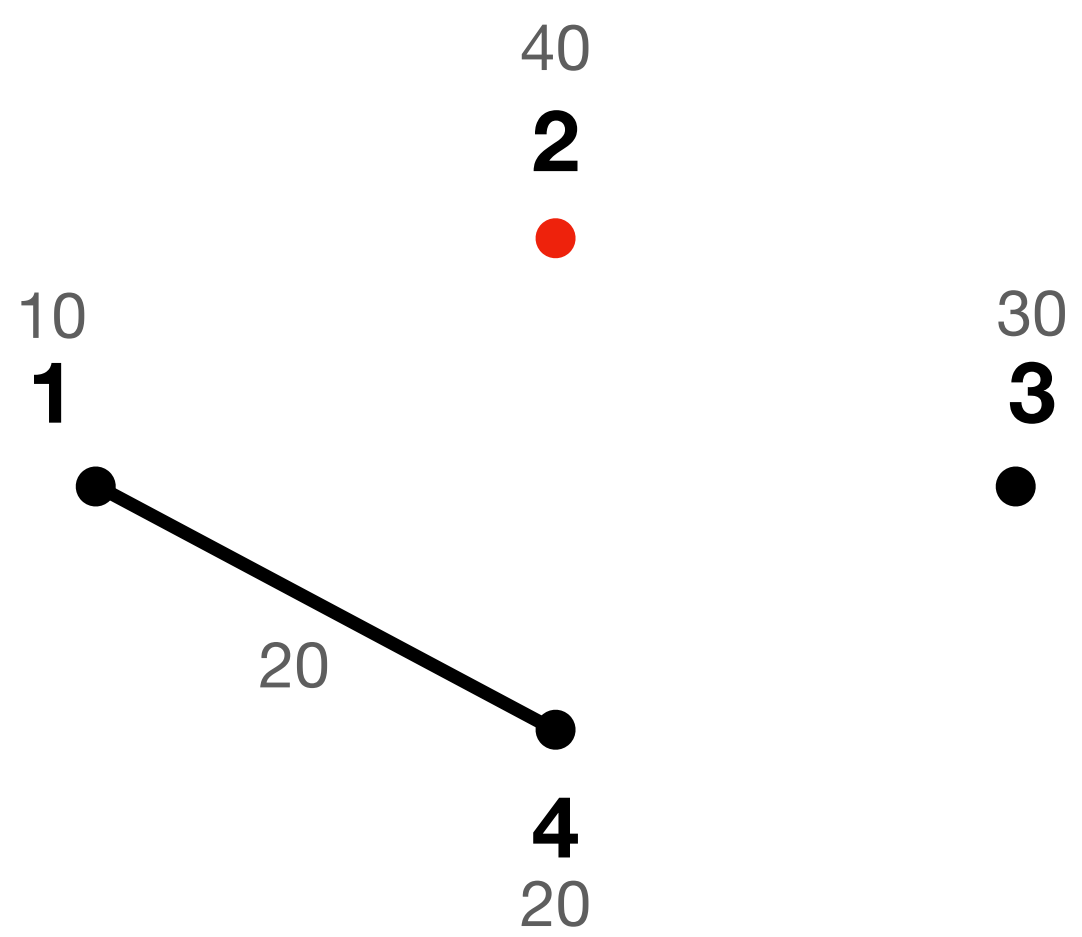
Boundary matrix reduction



	[1] 10	[4] 20		[3] 30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=30

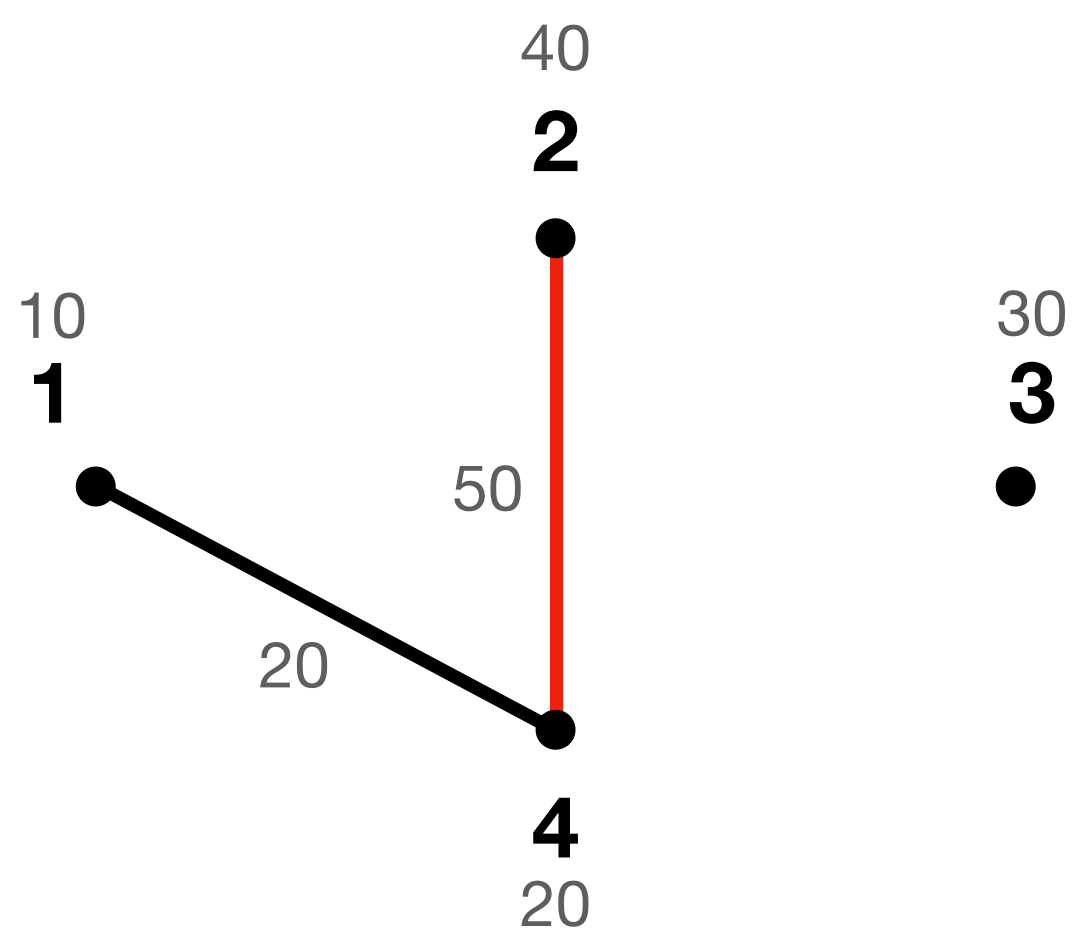
Boundary matrix reduction



	[1] 10	[4] 20		[3] 20	[2] 30					
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=40

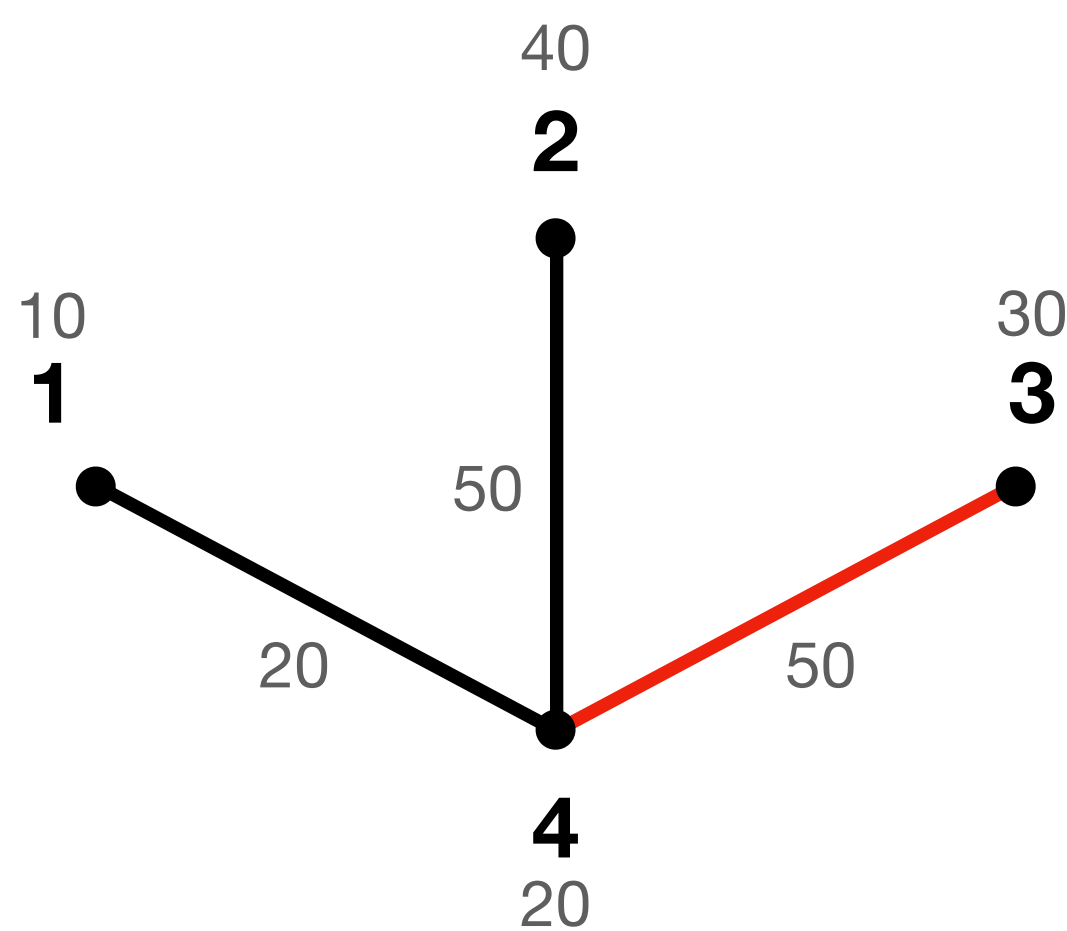
Boundary matrix reduction



	[1] 10	[4] 20		[3] 20	[2] 30					
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=50

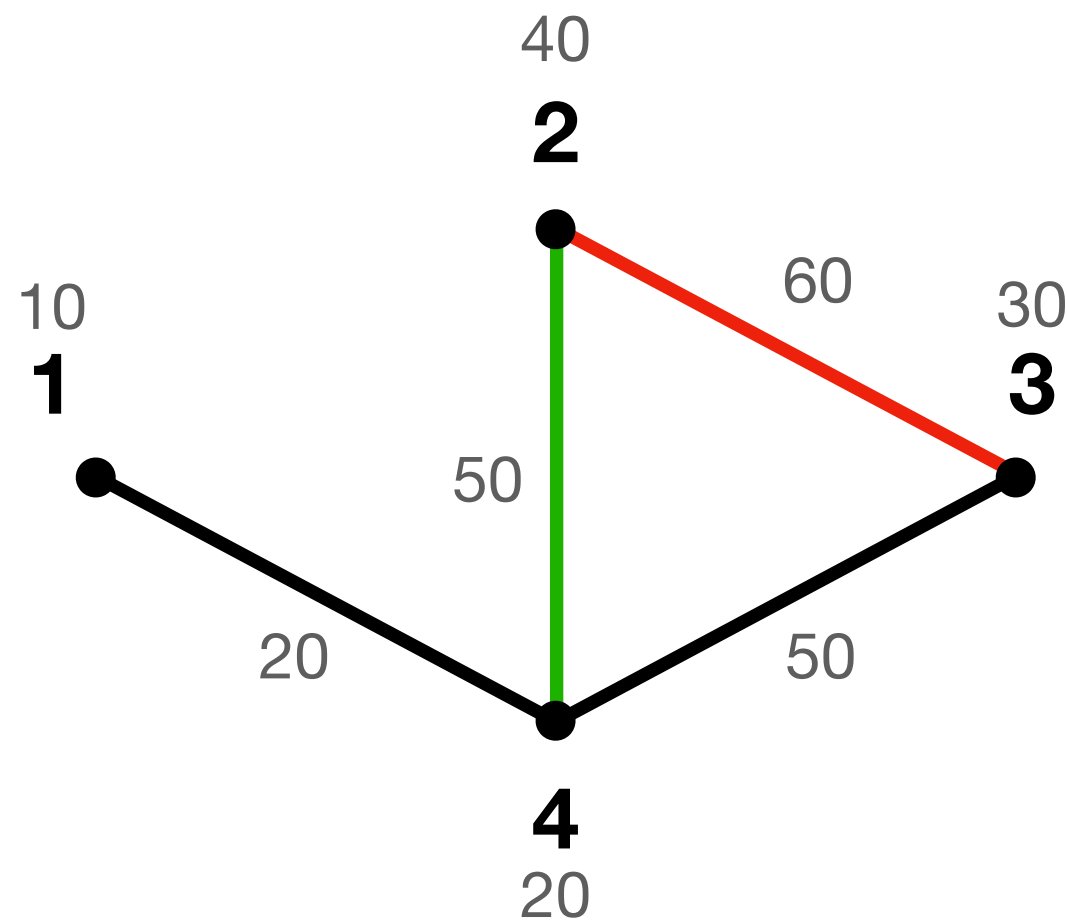
Boundary matrix reduction



	[1] 10	[4] 20		[3] 20	[2] 30					
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=50

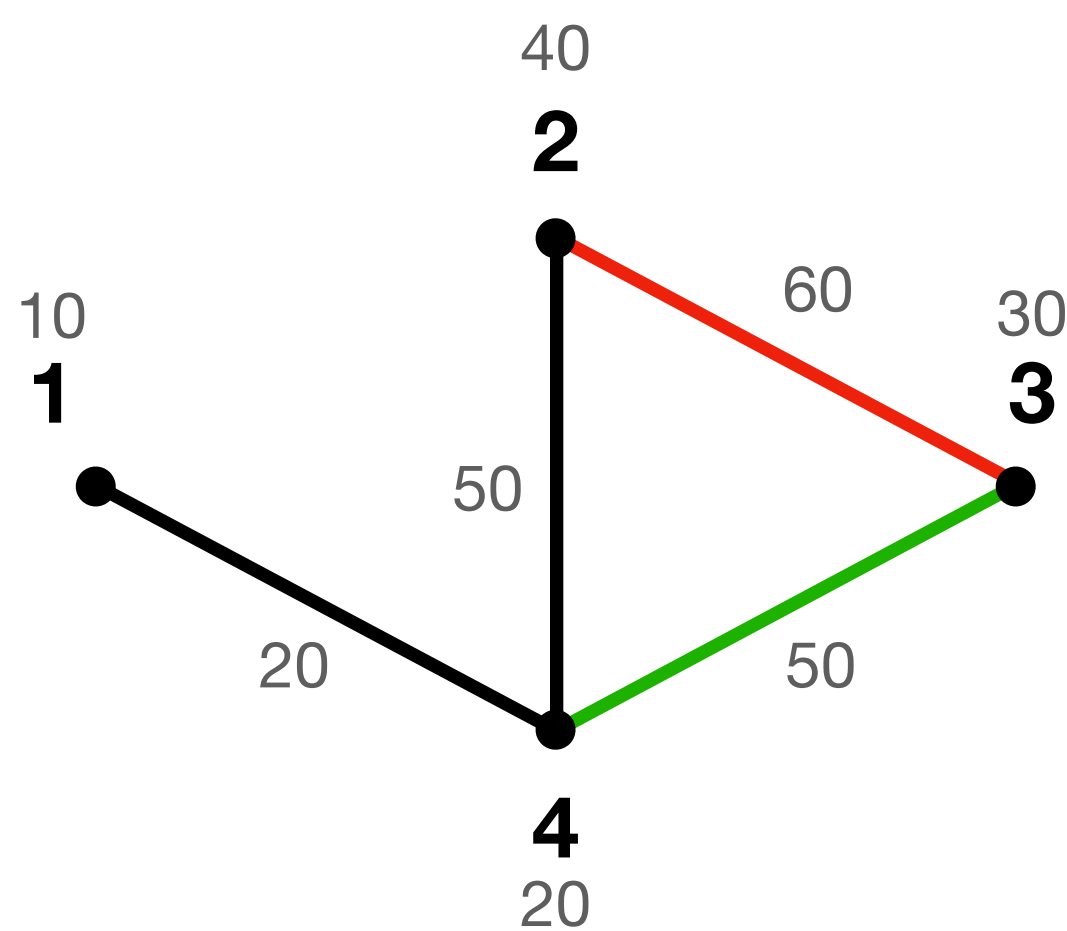
Boundary matrix reduction



	[1] 10	[4] 20		[3] 30	[2] 40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

t=60

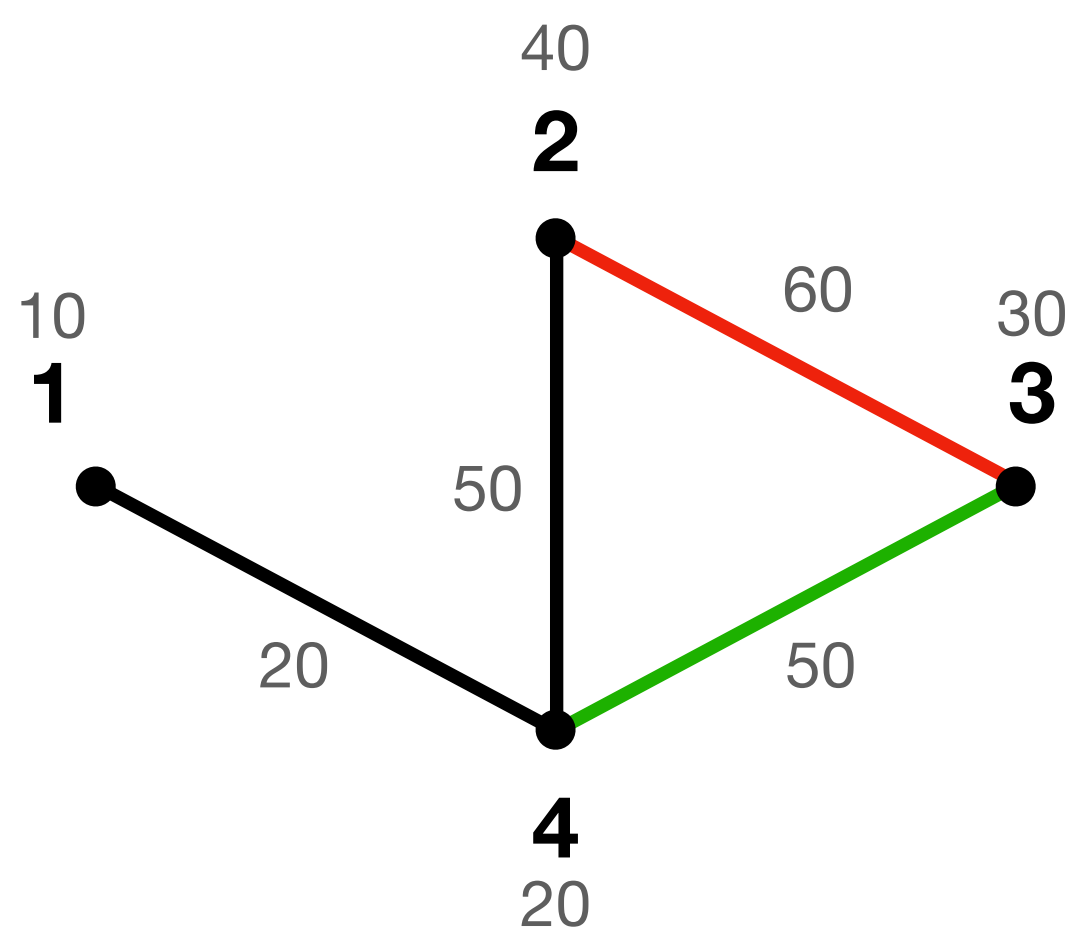
Boundary matrix reduction



	[1] 10	[4] 20	[14] 20	[3] 30	[2] 40	[24] 50	[34] 50	[23] 60	[23+24] 70	[12] 80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1	1		
14										
3							1	1		
2						1				1
24									1	
34									1	
23									1	
12										

t=60

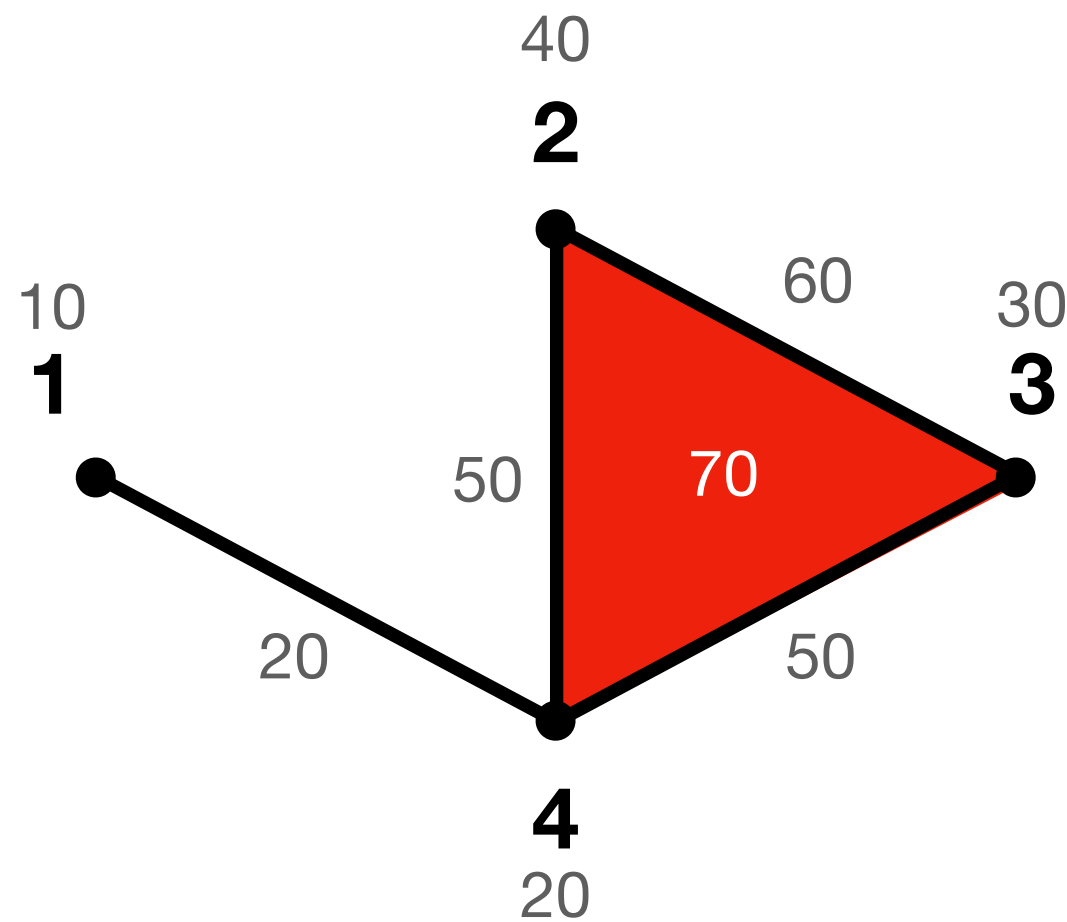
Boundary matrix reduction



	[1] 10	[4] 20	[14] 20	[3] 30	[2] 40	[24] 50	[34] 50	[23+24+34] 60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1			
2						1				1
24									1	
34									1	
23									1	
12										

t=60

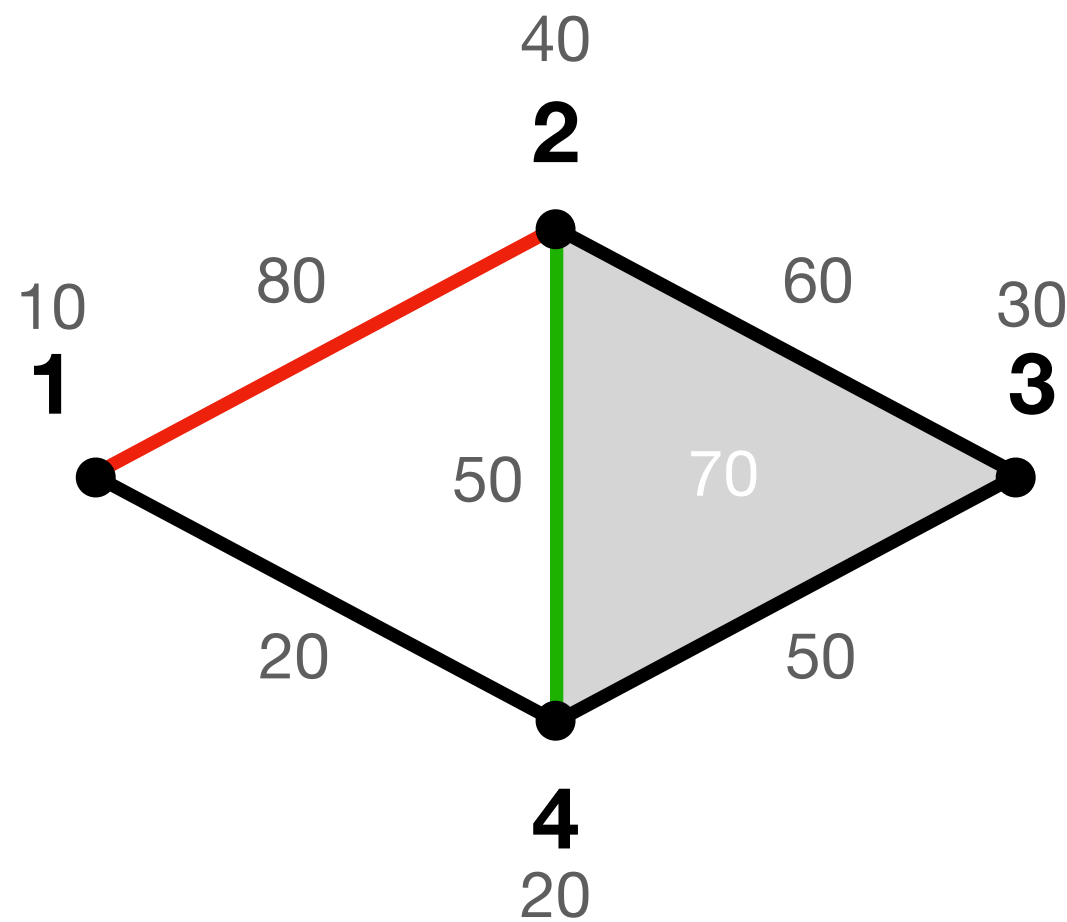
Boundary matrix reduction



	[1]	[4]		[3]	[2]		[23+24+34]			
	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1			
2						1				1
24									1	
34									1	
23									1	
12										

t=70

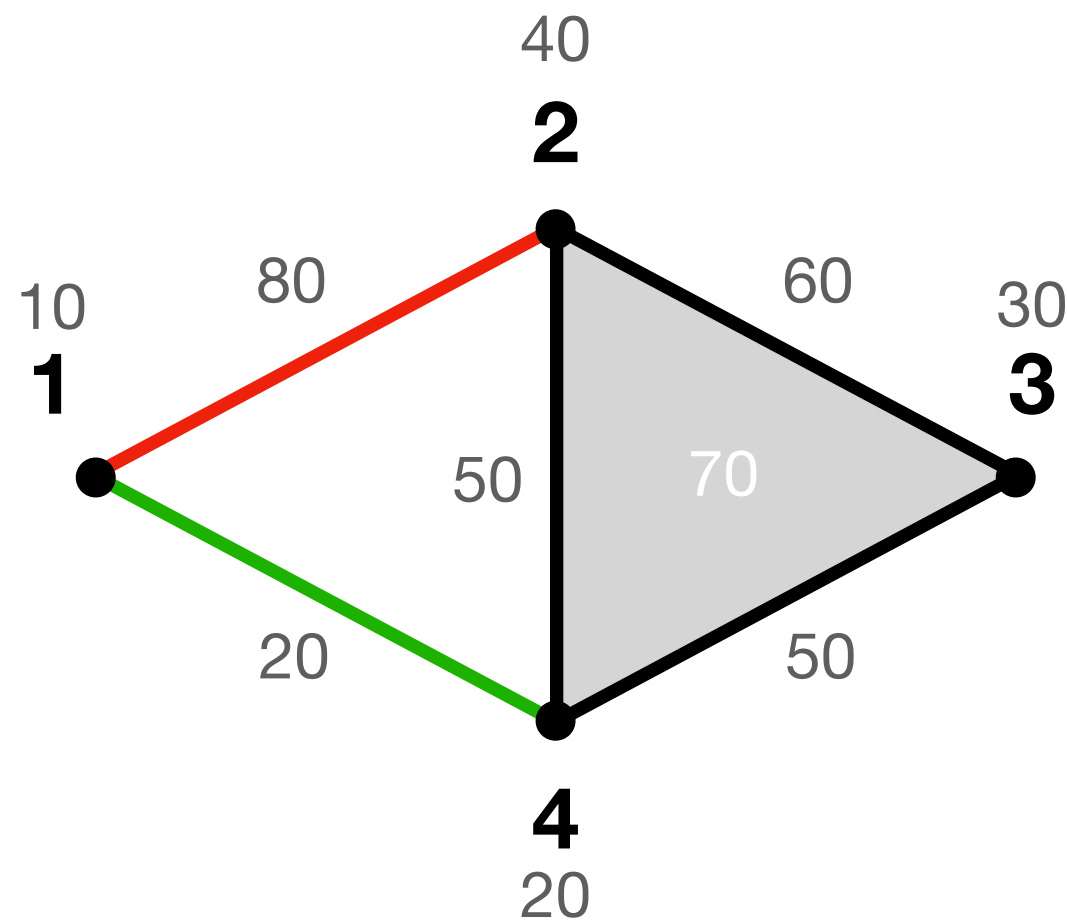
Boundary matrix reduction



	[1] 10	[4] 20		[3] 30	[2] 40		[23+24+34] 50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1			
2						1				1
24									1	
34									1	
23									1	
12										

t=80

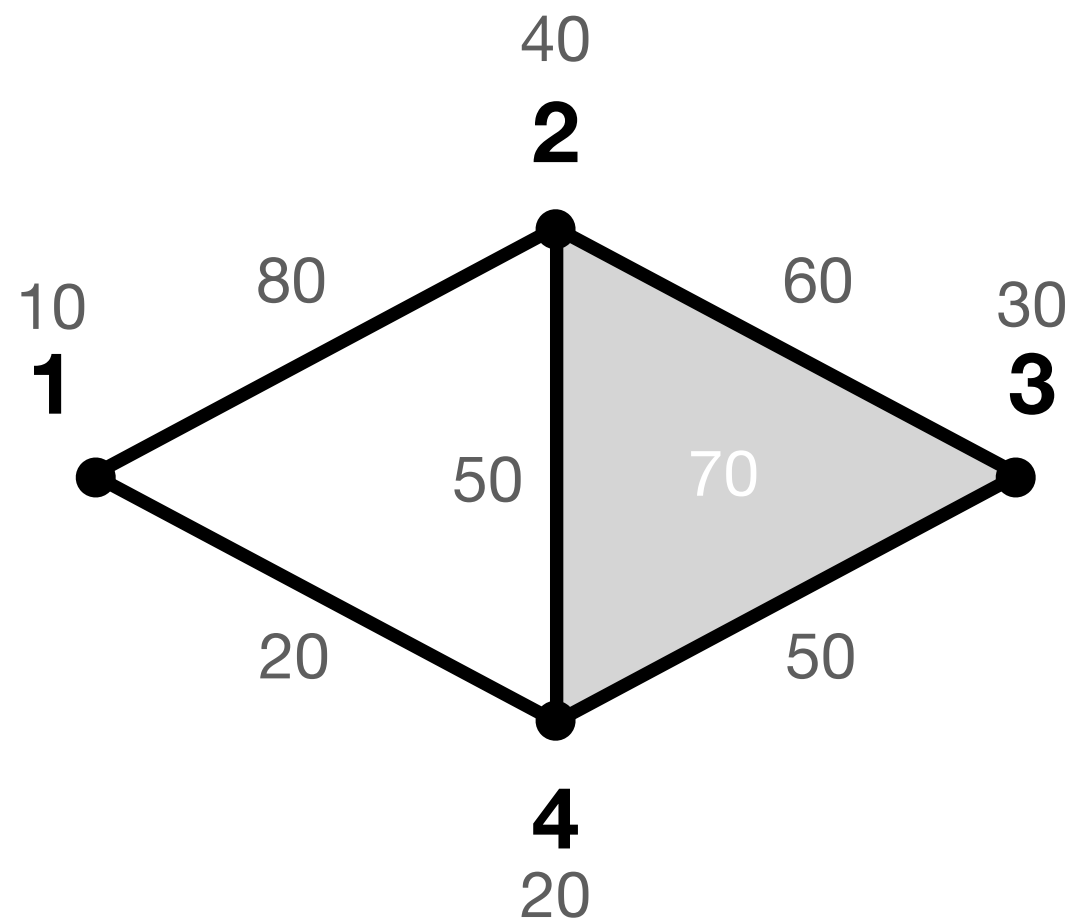
Boundary matrix reduction



	[1] 10	[4] 20	20	[3] 30	[2] 40	50	[23+24+34] 50 60 70			[12+24] 80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			1
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

t=80

Boundary matrix reduction

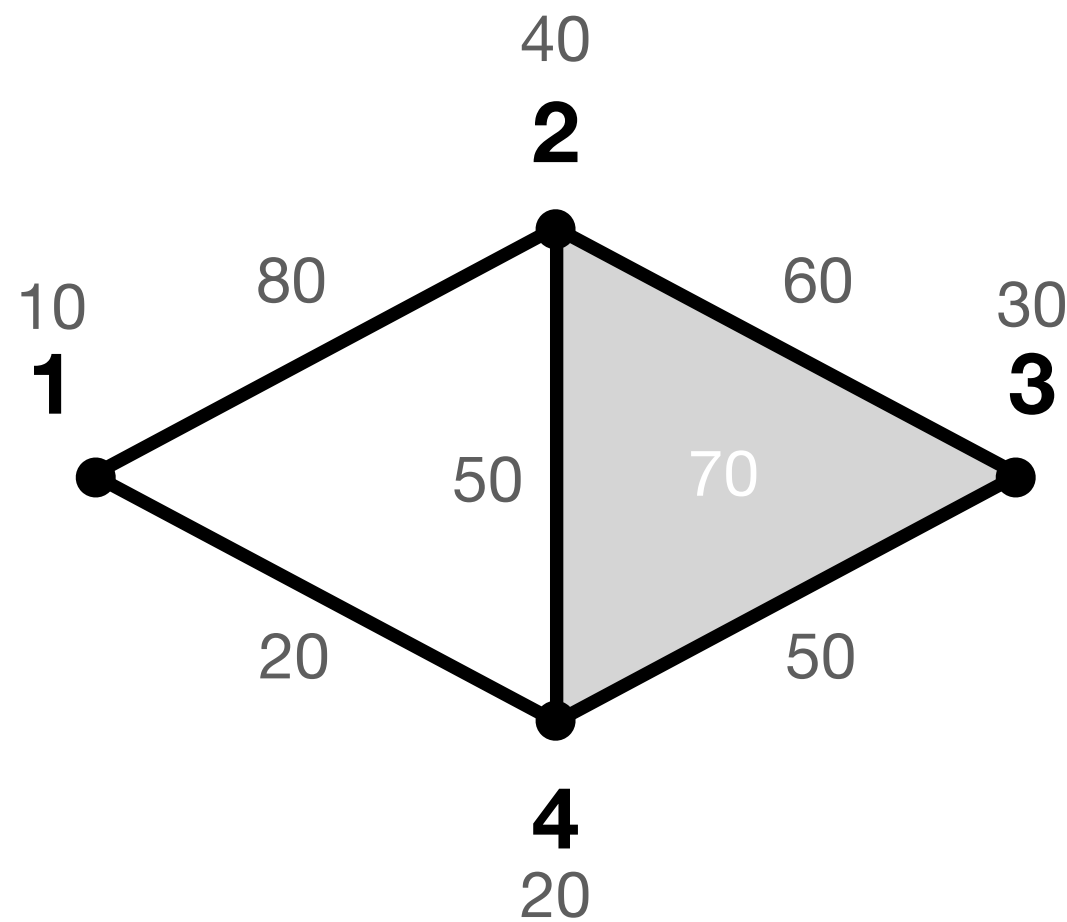


R =

	[1] 10	[4] 20	[14] 20	[3] 30	[2] 40	[24] 50	[34] 50	[23+24+34] 60	[12+14+24] 70	80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Matrix is called reduced if
all lowest nonzero elements are in unique rows

Boundary matrix reduction

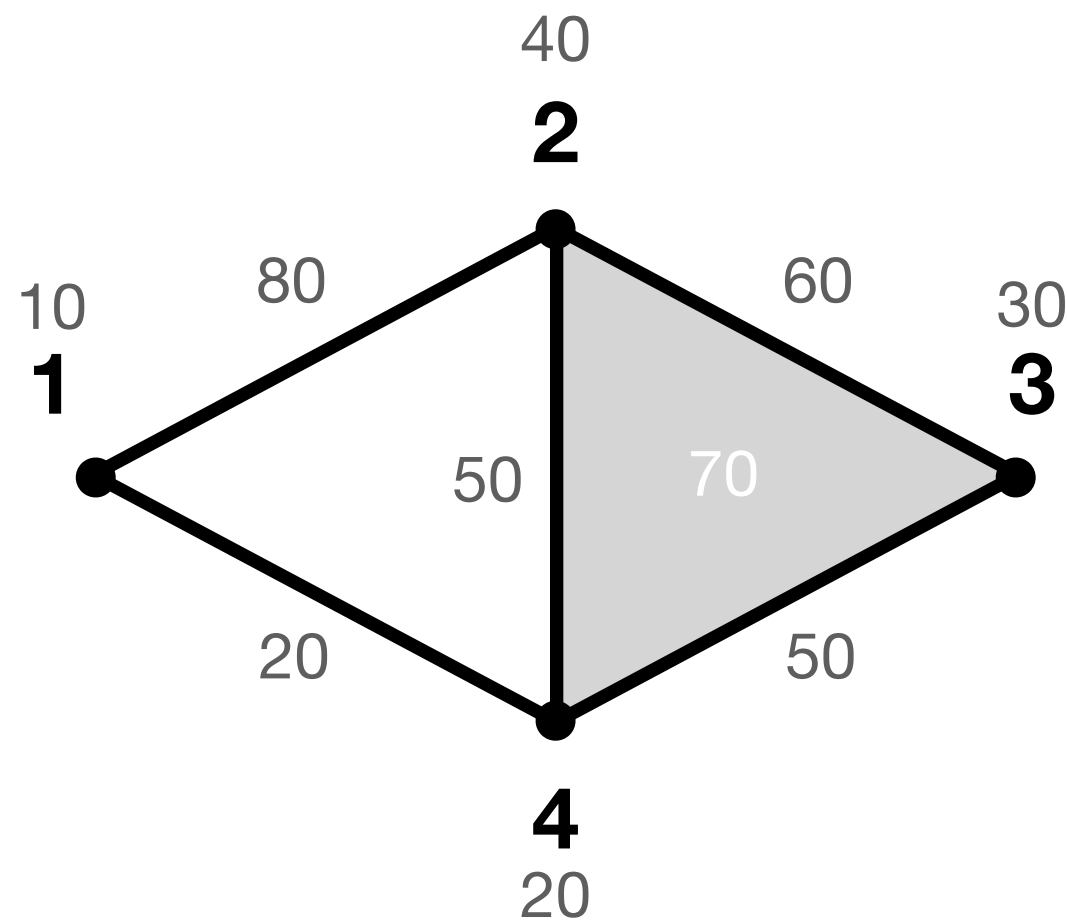


R =

	[1] 10	[4] 20		[3] 30	[2] 40			[23+24+34] 50	[12+14+24] 60	70	80
	1	4	14	3	2	24	34	23	234	12	
1			1								
4			1			1	1				
14											
3							1				
2						1					
24									1		
34									1		
23									1		
12											

Matrix is called reduced if
all lowest nonzero elements are in unique rows

Extracting information



R =

	[1] 10	[4] 20	[14] 20	[3] 30	[2] 40	[24] 50	[23+24+34] 50	[60]	[12+14+24] 70	[80]
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Persistence pairing

- (4, 14) 0

(2, 24) 0

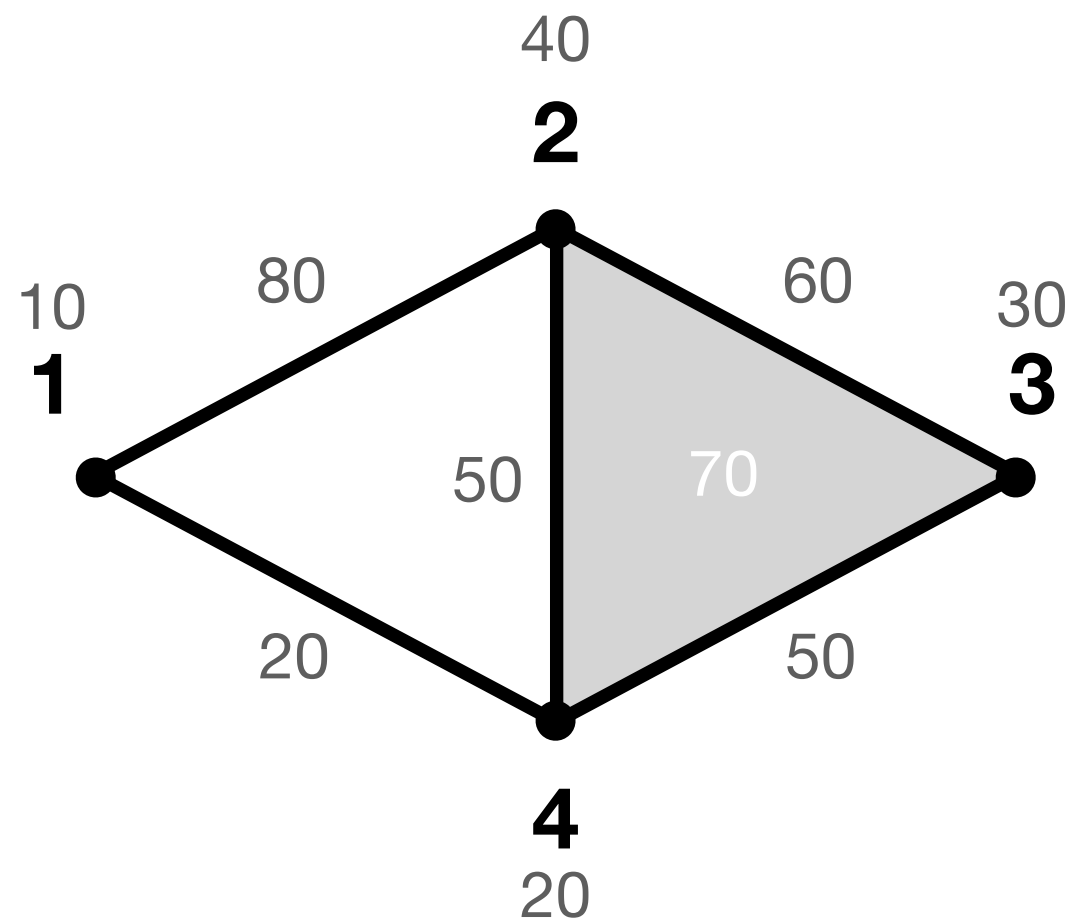
(3, 34) 0

(23, 234) 1
- (1, Ø) 0

(12, Ø) 1

Essential simplices correspond to unpaired empty columns

Extracting information



R =

	[1] 10	[4] 20	[14] 20	[3] 30	[2] 40	[24] 50	[23+24+34] 50	[60] 60	[12+14+24] 70	[80] 80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Persistence pairing

- (4, 14) 0

(2, 24) 0

(3, 34) 0

(23, 234) 1
- (1, Ø) 0

(12, Ø) 1

Persistence diagram

- (20, 20) 0

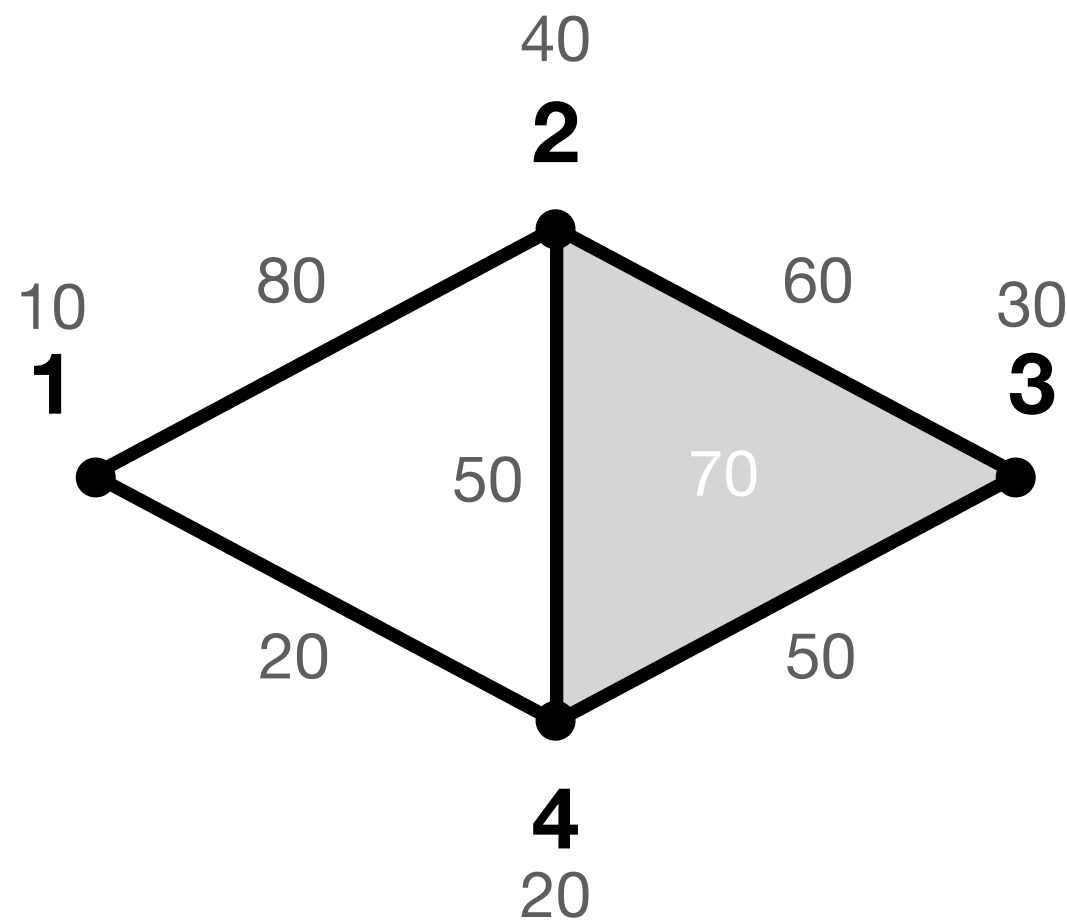
(40, 50) 0

(30, 50) 0

(60, 70) 1
- (10, Ø) 0

(80, Ø) 1

Extracting information



R =

	[1] 10	[4] 20	[14] 20	[3] 30	[2] 40	[24] 50	[23+24+34] 50	[12+14+24] 60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

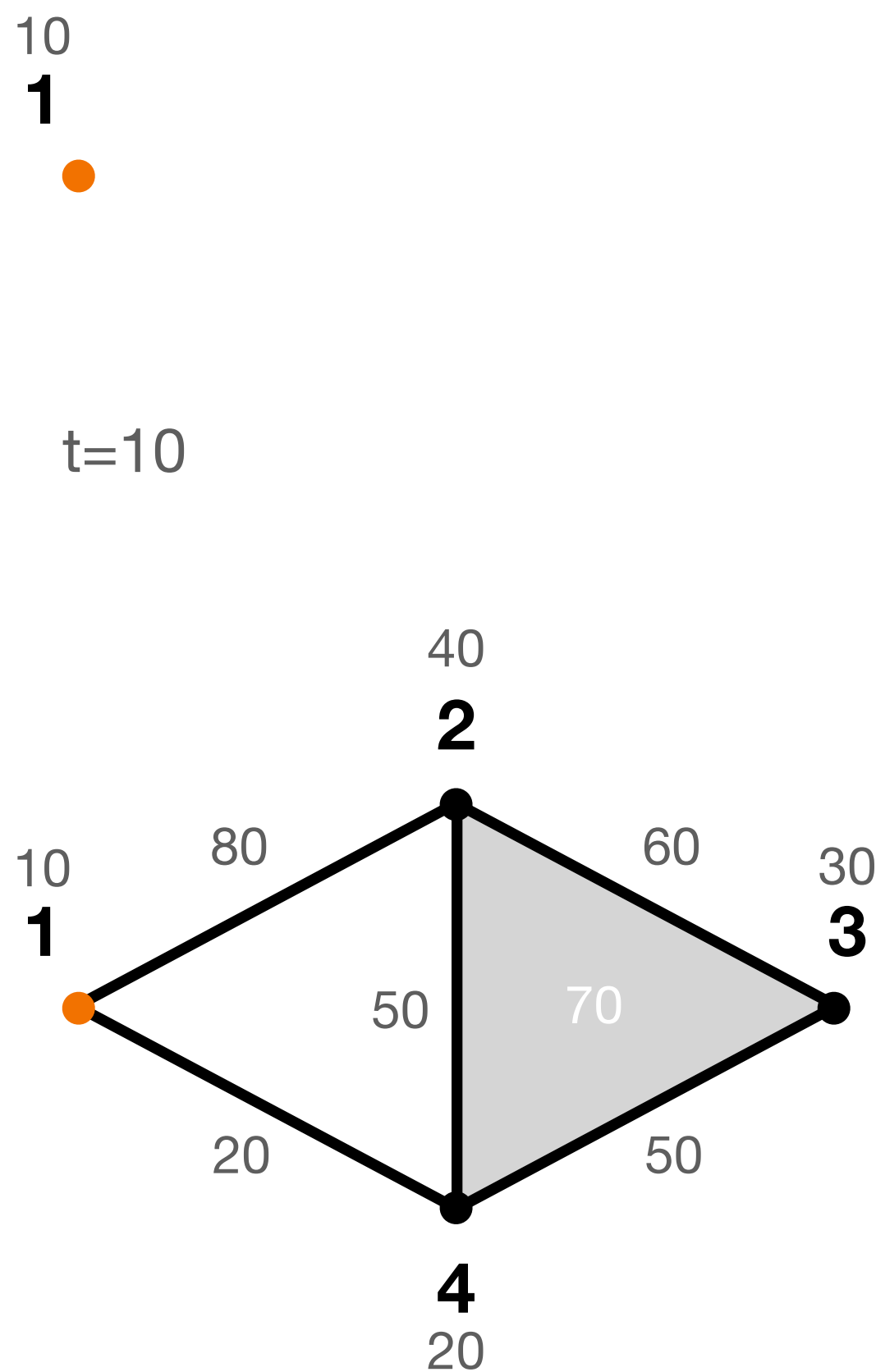
Persistence pairing [representatives]

(4, 14) 0 [4] (1, ∅) 0 [1]
(2, 24) 0 [2] (12, ∅) 1 [12+14+24]
(3, 34) 0 [3]
(23, 234) 1 [23+24+34]

Persistence diagram

(20, 20) 0 (10, ∞) 0
(40, 50) 0 (80, ∞) 1
(30, 50) 0
(60, 70) 1

Representatives

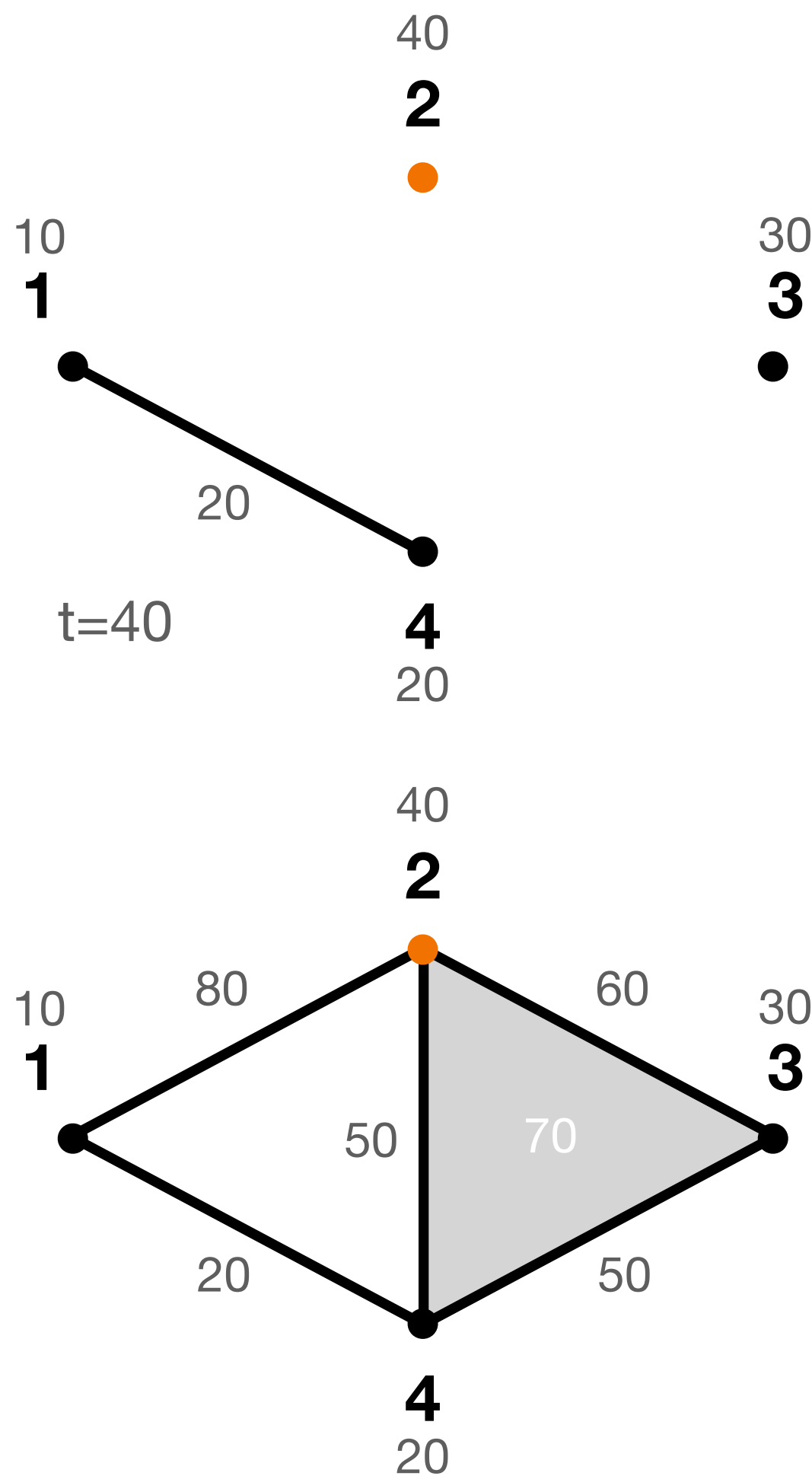


R =

	[1]	[4]		[3]	[2]		[23+24+34]	[12+14+24]		
	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Representatives are given by the linear combination of columns
corresponding to the reduced columns

Representatives

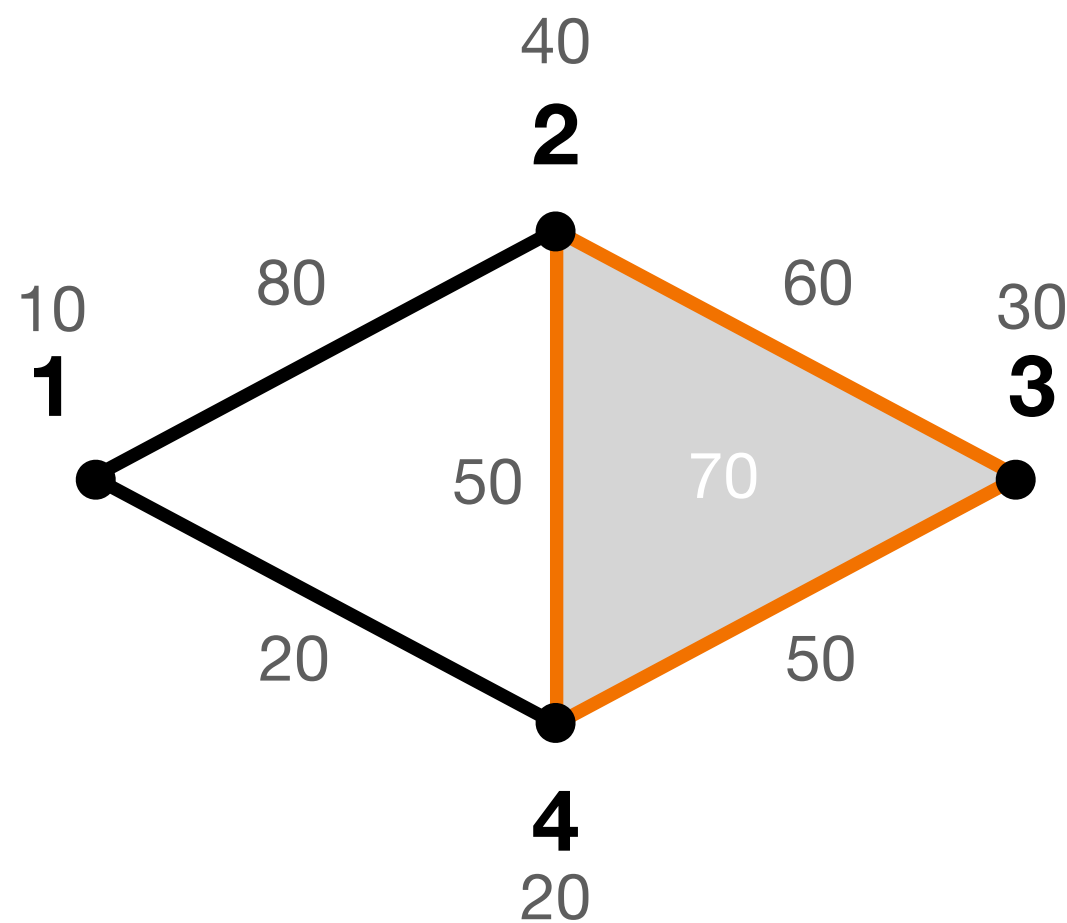
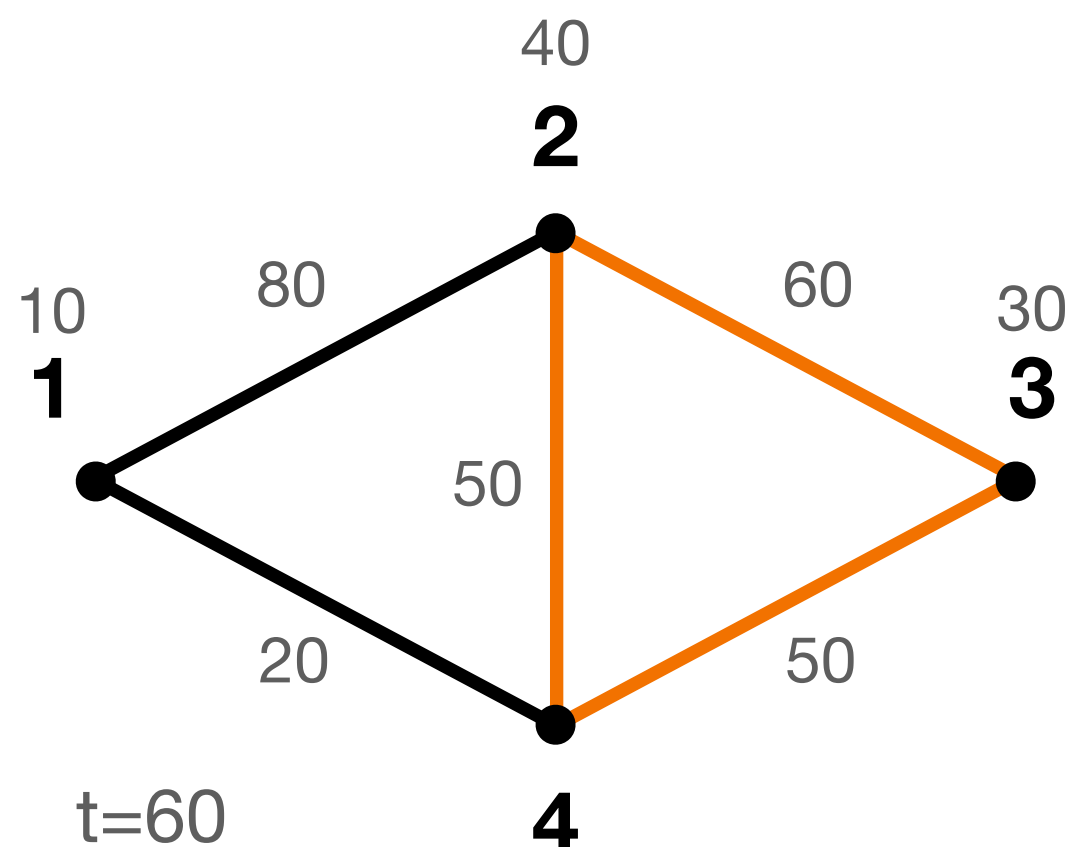


R =

	[1] 10	[4] 20	[3] 20	[2] 30	[2] 40	[23+24+34] 50	[23+24+34] 50	[12+14+24] 60	[12+14+24] 70	[12+14+24] 80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Representatives are given by the linear combination of columns
corresponding to the reduced columns

Representatives

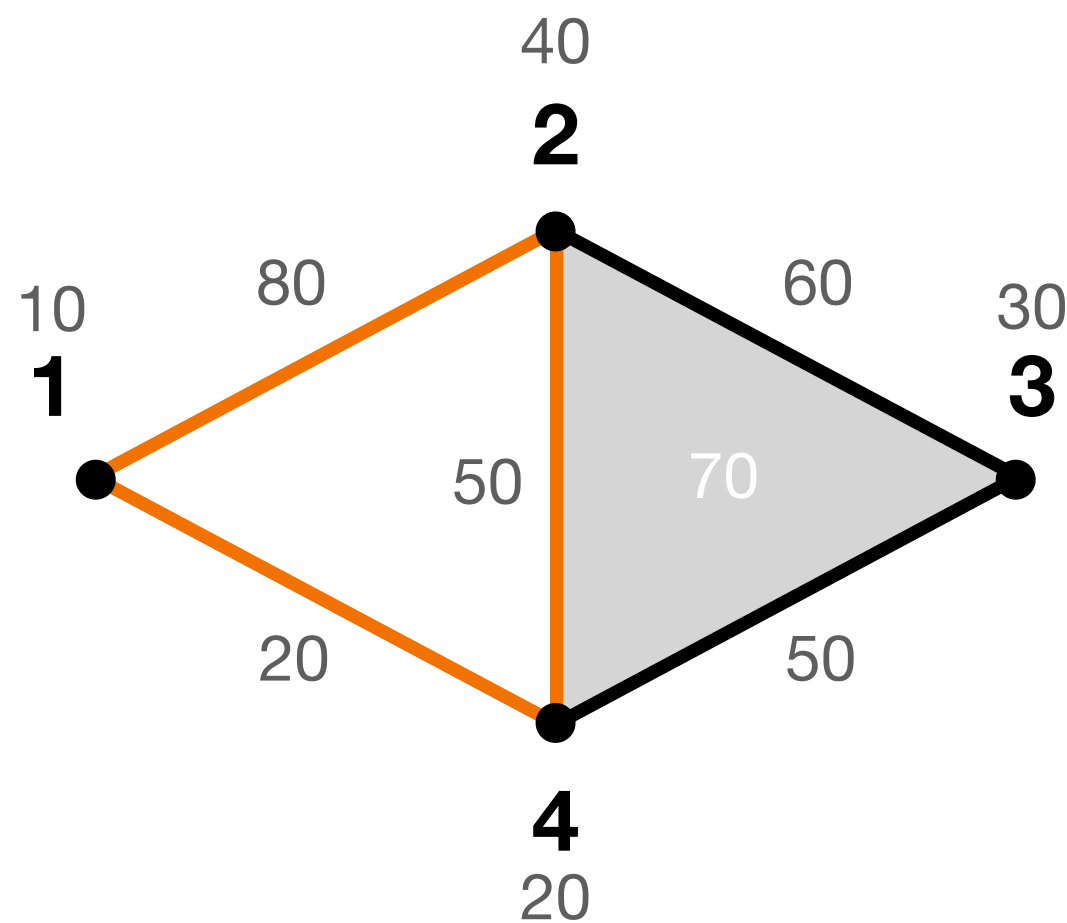
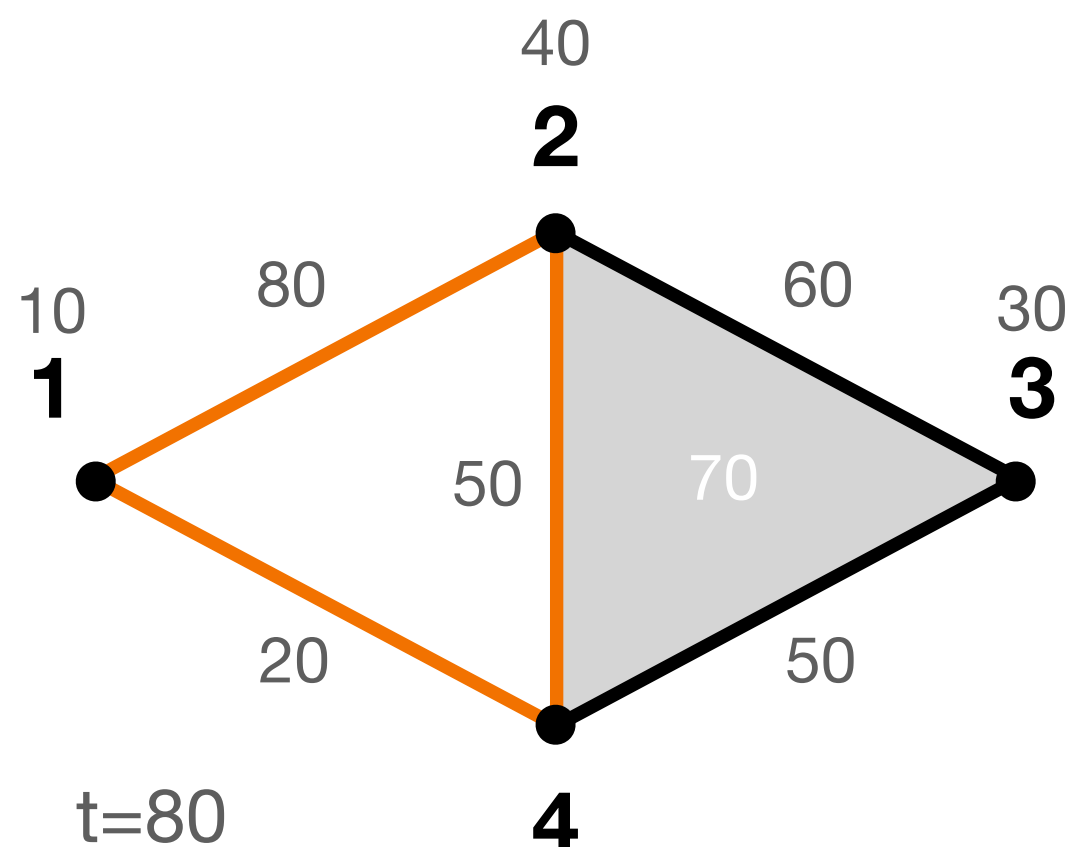


R =

	[1] 10	[4] 20	[14] 20	[3] 30	[2] 40	[24] 50	[34] 50	[23+24+34] 60	[12+14+24] 70	[12] 80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Representatives are given by the linear combination of columns corresponding to the reduced columns

Representatives

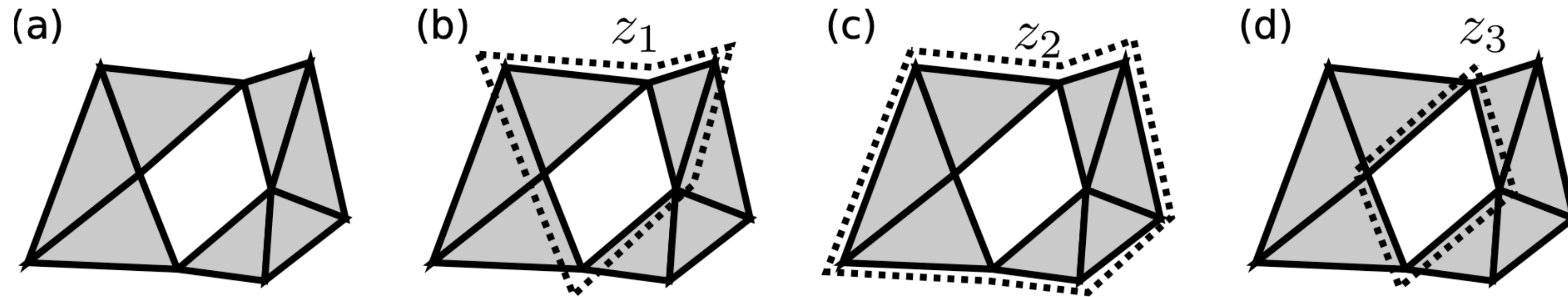


R =

	[1] 10	[4] 20	[3] 20	[2] 30		[23+24+34] 50		[12+14+24] 50		
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Representatives are given by the linear combination of columns corresponding to the reduced columns

Homologous cycles

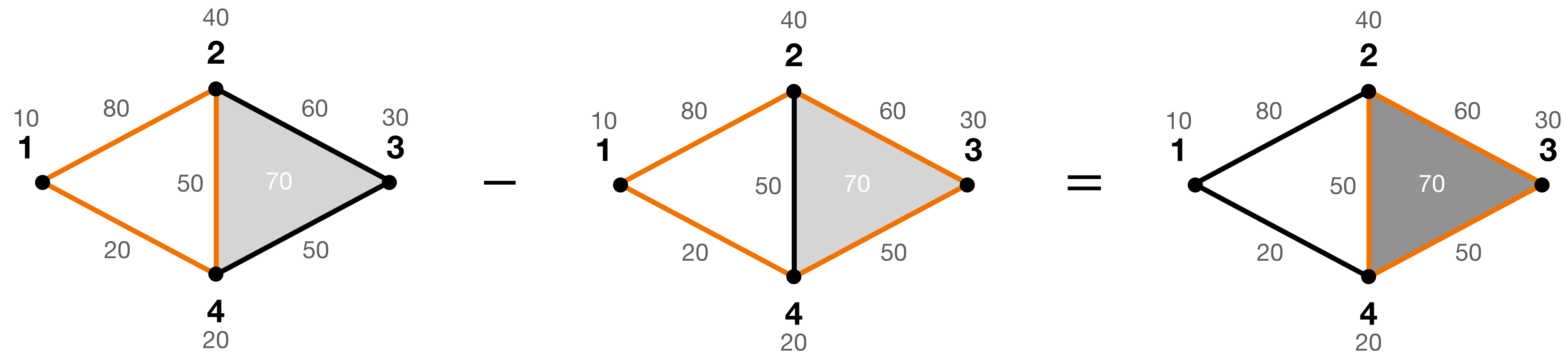


Homologous cycles

$$z \sim z' \iff z - z' \in B_k$$

$$B_k = \{c \in C_k \mid \partial_{k+1}d = c, \text{ for some } d \in C_{k+1}\}$$

Homologous cycles



Homologous cycles

$$z \sim z' \iff z - z' \in B_k$$

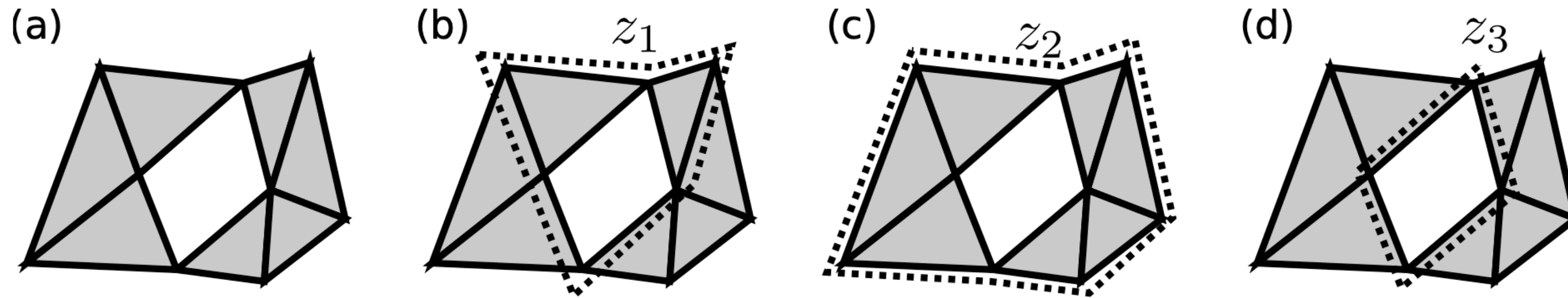
$$[12 + 14 + 24] - [12 + 14 + 23 + 34] = [23 + 24 + 34]$$

$$\partial_2([234]) = [23 + 24 + 34]$$

$$B_k = \{c \in C_k \mid \partial_{k+1}d = c, \text{ for some } d \in C_{k+1}\}$$

Representatives

Optimal representatives, one hole



Optimal cycle

minimize $\|z\|_0$ subject to $z \sim z_1$.

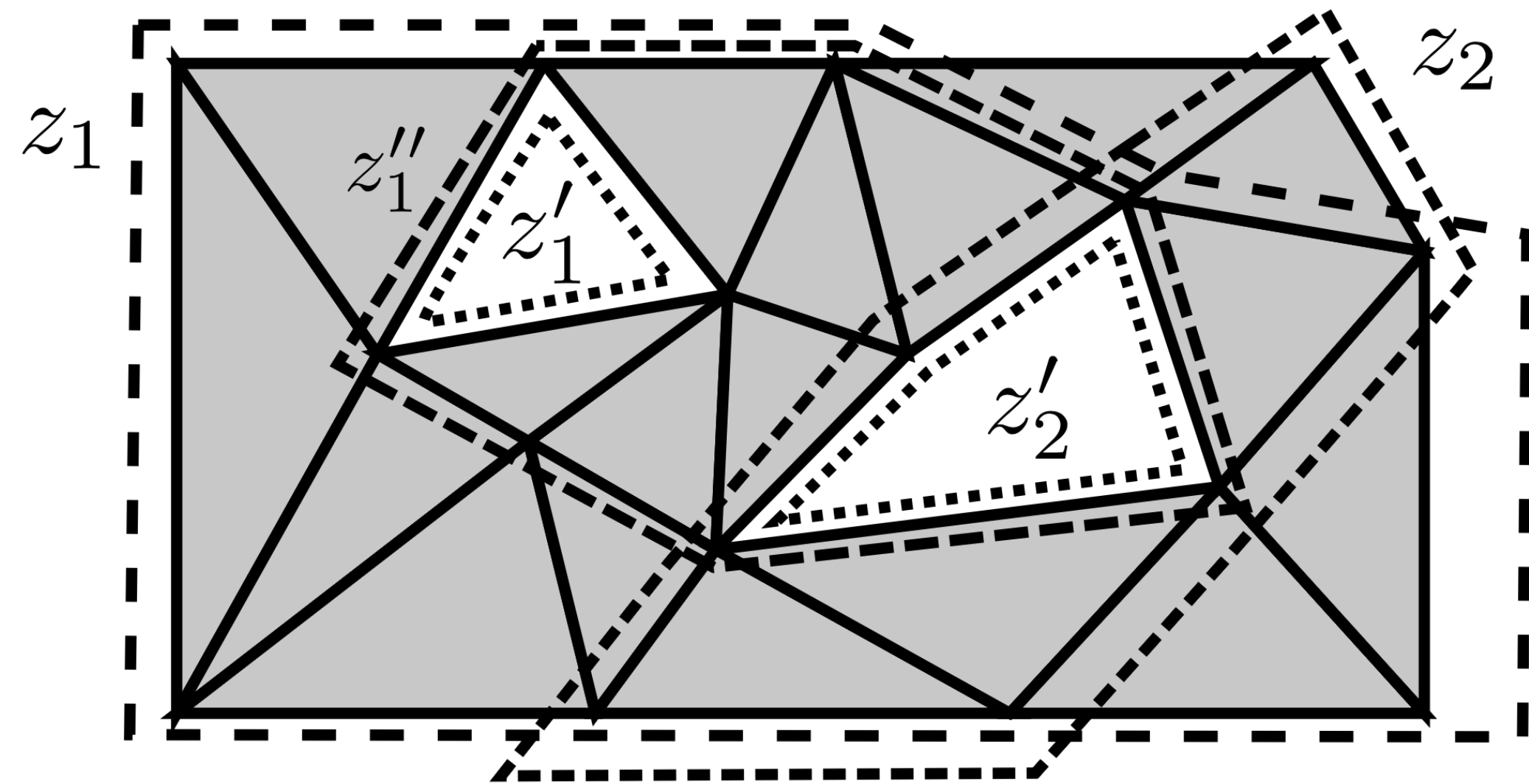
minimize $\|z\|_0$ subject to

$$z = z_1 + \partial w,$$

$$w \in C_2(X).$$

Representatives

Optimal representatives, several holes



Optimal cycle

minimize $\|z\|_0$, subject to:

$$z = z_1 + \partial w + kz_2,$$

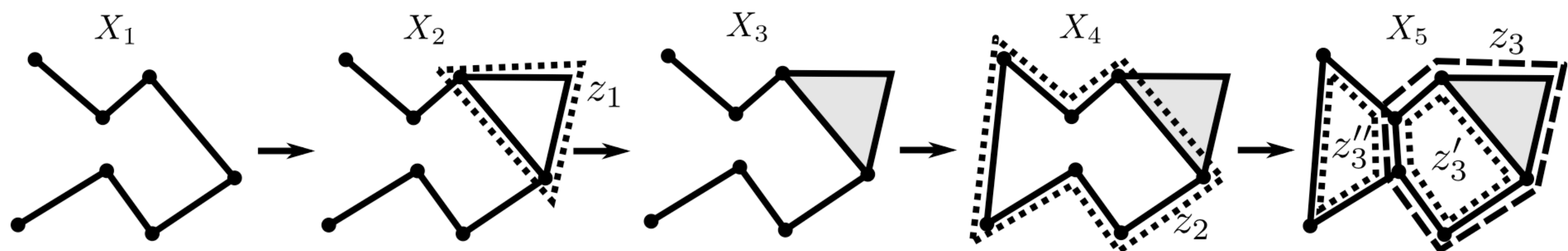
$$w \in C_2(X),$$

$$k \in \mathbb{Z}_2.$$

“Fill” a hole by considering $Z_1 / (B_1 \oplus \langle z_2' \rangle)$ instead of Z_1 / B_1

Persistence representatives

Optimal representatives w.r.t. a filtration — cycles



- z1 born at X2
 dead at X3
- z2 born at X4
- z3 born at X5

Algorithm 1 Computing an optimal cycle on a filtration.

Compute $D_q(\mathbb{X})$ and persistence cycles $z_1, \dots, z_n$
 Have $(b_i, d_i) \in D_q(\mathbb{X})$ be chosen by a user
 Solve the following optimization problem:

$$\begin{aligned} &\text{minimize } \|z\|_1 \quad \text{subject to} \\ & \quad z = z_i + \partial w + \sum_{j \in T_i} \alpha_j z_j, \\ & \quad w \in C_{q+1}(X_{b_i}), \\ & \quad \alpha_j \in \mathbb{k}, \\ & \quad \text{where } T_i = \{j \mid b_j < b_i < d_j\} \end{aligned}$$

Find $z^* \sim z_3$
 minimize $\|z\|_0$ subject to
 $z = z_3 + \partial w + k z_2,$
 $w \in C_2(X_5),$
 $k \in \mathbb{k}.$

Representatives

Scaffolds

Given a graph G a homological scaffold $H(G)$ is a subgraph of G induced by the edges present in the representatives of homology classes.

The *frequency homological scaffold* $H^F(G)$ is the network composed of all the cycle paths corresponding to generators, where an edge e is weighted by the number of different cycles it belongs to.

The *persistence homological scaffold* $H^P(G)$ is the network composed of all the cycle paths corresponding to generators weighted by their persistence.

If an edge e belongs to multiple cycles $z_0, z_1, \dots, z_s$, its weight is defined as the sum of the generators' persistence.

$$\omega_e^P = \sum_{[z]_i} \mathbf{1}_{e \in [z]_i}$$

$$\omega_e^F = \sum_{[z]_i \mid e \in [z]_i} \pi_{[z]_i}$$

Harmonic representatives

Hodge Laplacian operator

$$C_2 \begin{array}{c} \xrightarrow{\partial_2} \\ \xleftarrow{\partial_2^*} \end{array} C_1 \begin{array}{c} \xrightarrow{\partial_1} \\ \xleftarrow{\partial_1^*} \end{array} C_0$$

$$\mathbf{L}_k = \mathbf{B}_k^T \mathbf{B}_k + \mathbf{B}_{k+1} \mathbf{B}_{k+1}^T$$

Betti numbers via Hodge Laplacian

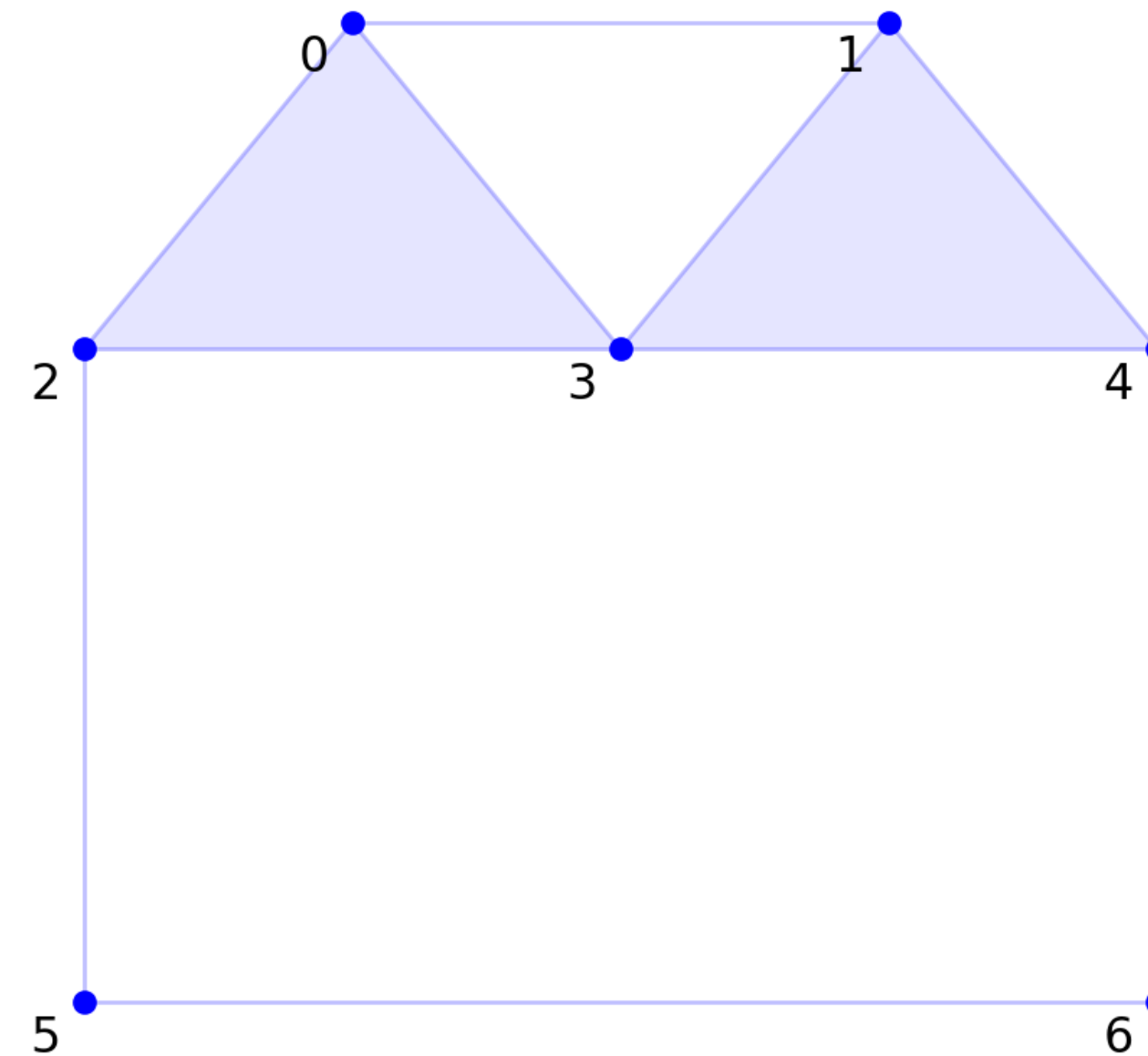
$$\beta_k = \dim \ker(\mathbf{L}_k)$$

Harmonic cycles

$$z_k^H \in \ker L_k$$

Hodge decomposition

$$C_k = \text{im} \partial_{k+1} \oplus \ker L_k \oplus \partial_k^T$$



Harmonic representatives

Hodge Laplacian operator

$$C_2 \begin{matrix} \xrightarrow{\partial_2} \\ \xleftarrow{\partial_2^*} \end{matrix} C_1 \begin{matrix} \xrightarrow{\partial_1} \\ \xleftarrow{\partial_1^*} \end{matrix} C_0$$

$$\mathbf{L}_k = \mathbf{B}_k^T \mathbf{B}_k + \mathbf{B}_{k+1} \mathbf{B}_{k+1}^T$$

Betti numbers via Hodge Laplacian

$$\beta_k = \dim \ker(\mathbf{L}_k)$$

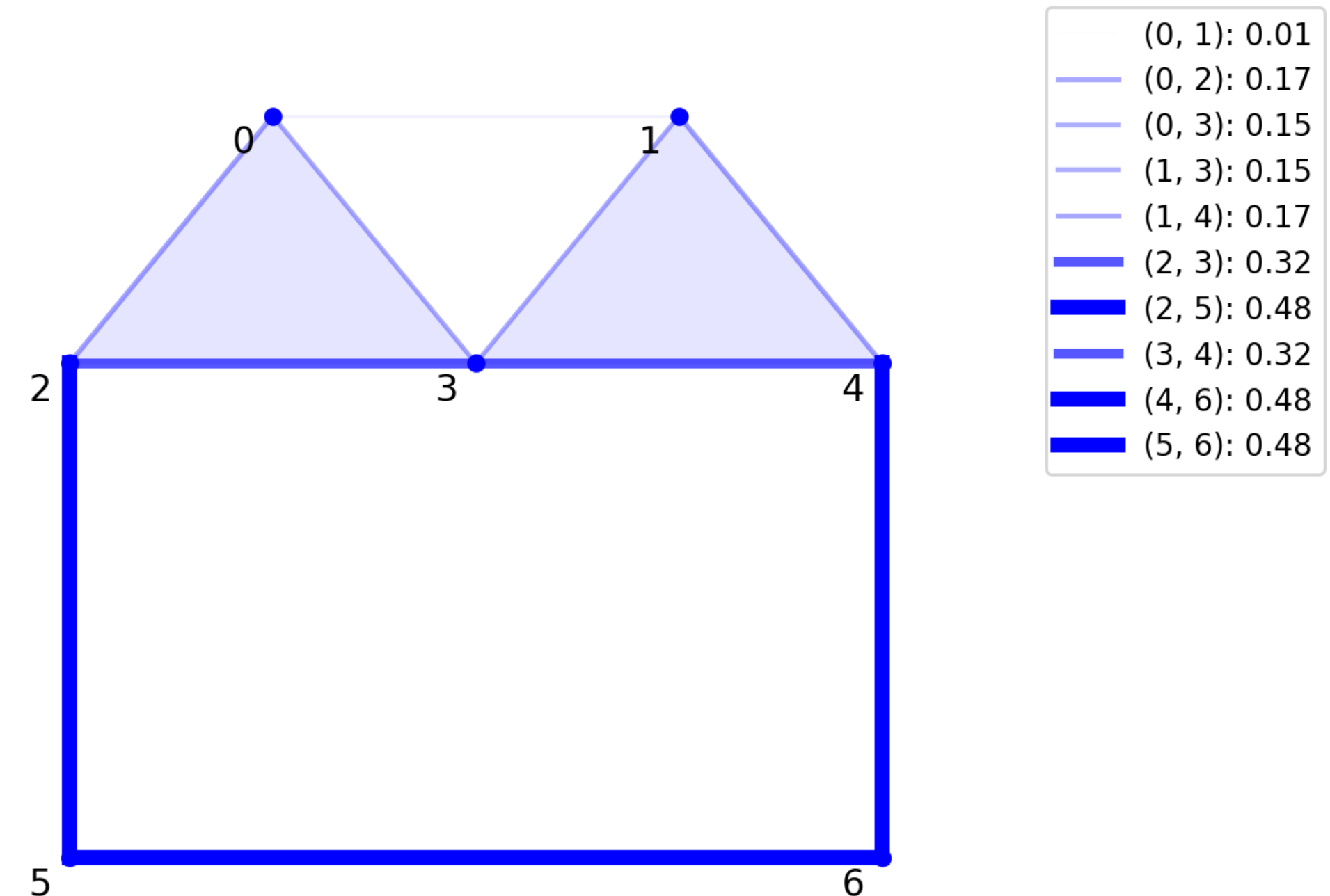
Harmonic cycles

$$z_k^H \in \ker L_k$$

Eigenvectors corresponding to zero eigenvalues of $\mathbf{L}_k$

Hodge decomposition

$$C_k = \text{im} \partial_{k+1} \oplus \ker L_k \oplus \partial_k^T$$



Harmonic representatives

Hodge Laplacian operator

$$C_2 \begin{matrix} \xrightarrow{\partial_2} \\ \xleftarrow{\partial_2^*} \end{matrix} C_1 \begin{matrix} \xrightarrow{\partial_1} \\ \xleftarrow{\partial_1^*} \end{matrix} C_0$$

$$\mathbf{L}_k = \mathbf{B}_k^T \mathbf{B}_k + \mathbf{B}_{k+1} \mathbf{B}_{k+1}^T$$

Betti numbers via Hodge Laplacian

$$\beta_k = \dim \ker(\mathbf{L}_k)$$

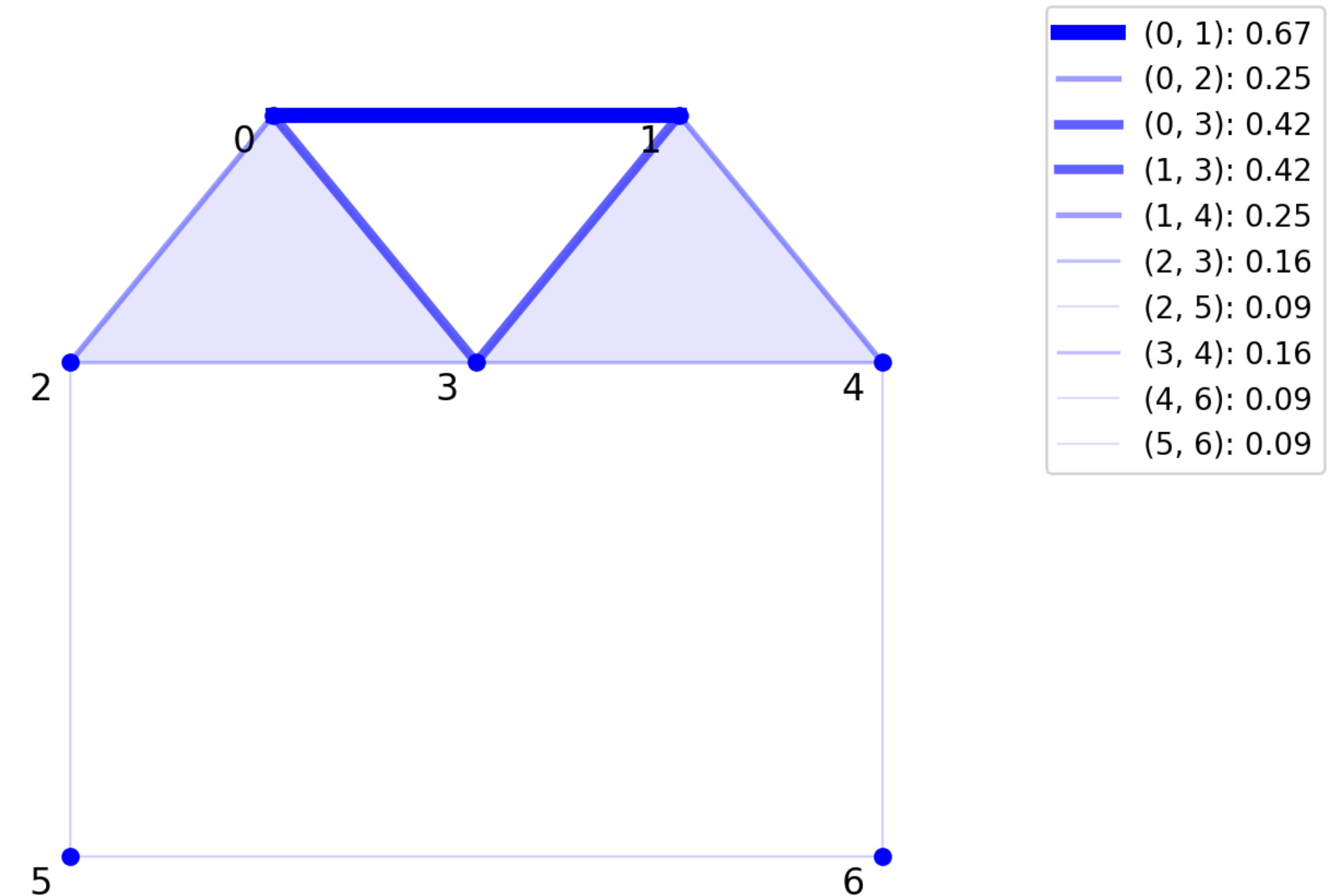
Harmonic cycles

$$z_k^H \in \ker L_k$$

Eigenvectors corresponding to zero eigenvalues of $\mathbf{L}_k$

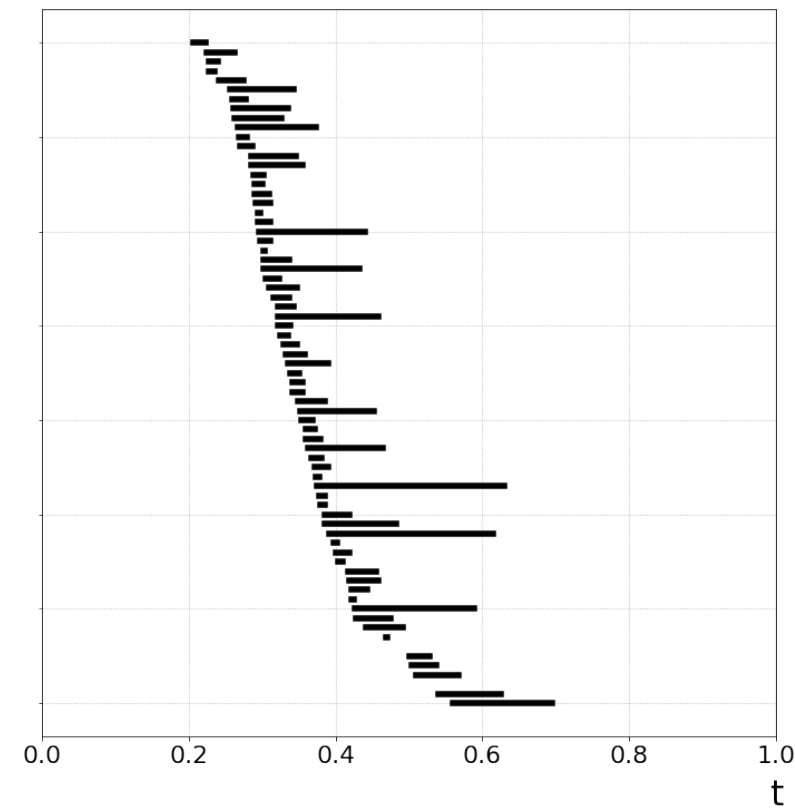
Hodge decomposition

$$C_k = \text{im} \partial_{k+1} \oplus \ker L_k \oplus \partial_k^T$$

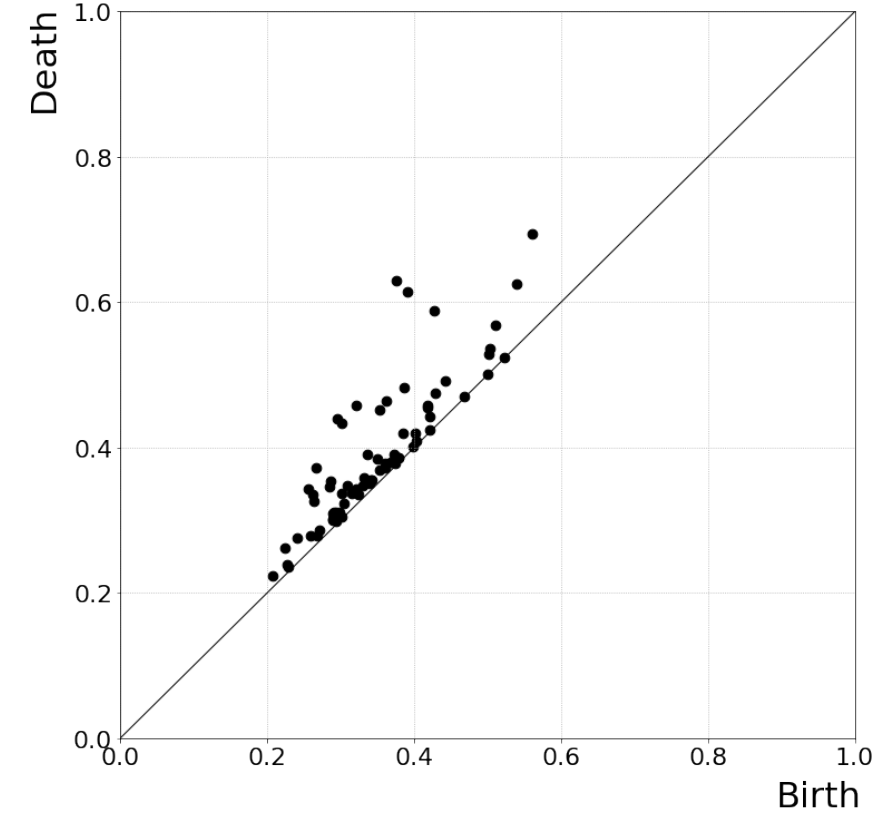


Persistent harmonic representatives

Persistence Barcode



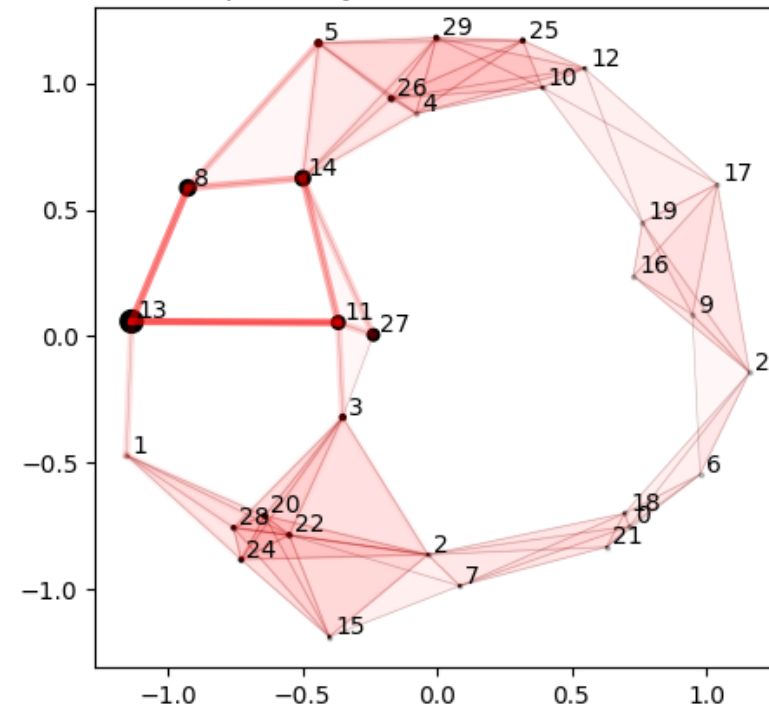
Persistence Diagram



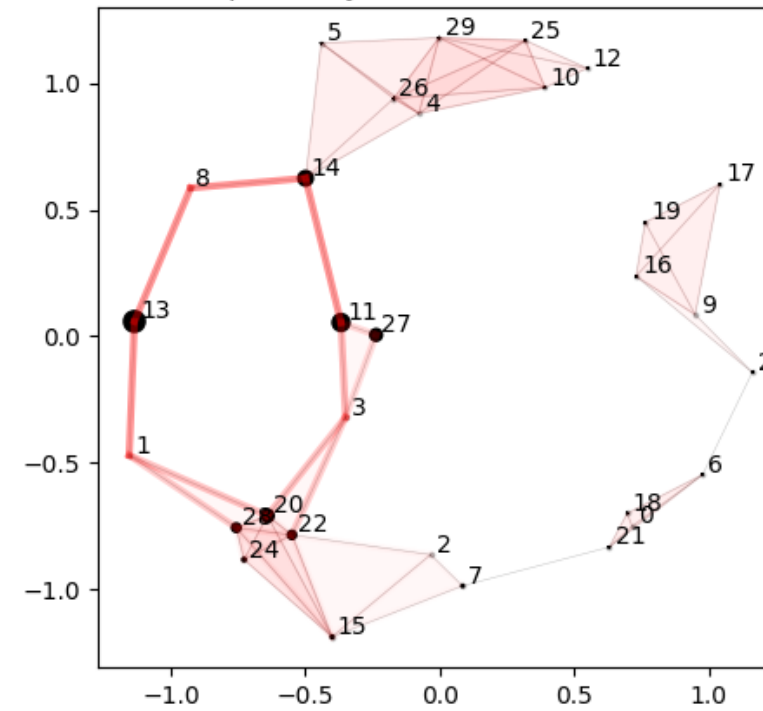
Persistence diagram

$$D_{f_\theta}(X) = \{(f(b_i), f(d_i))\}_{i \in I}$$

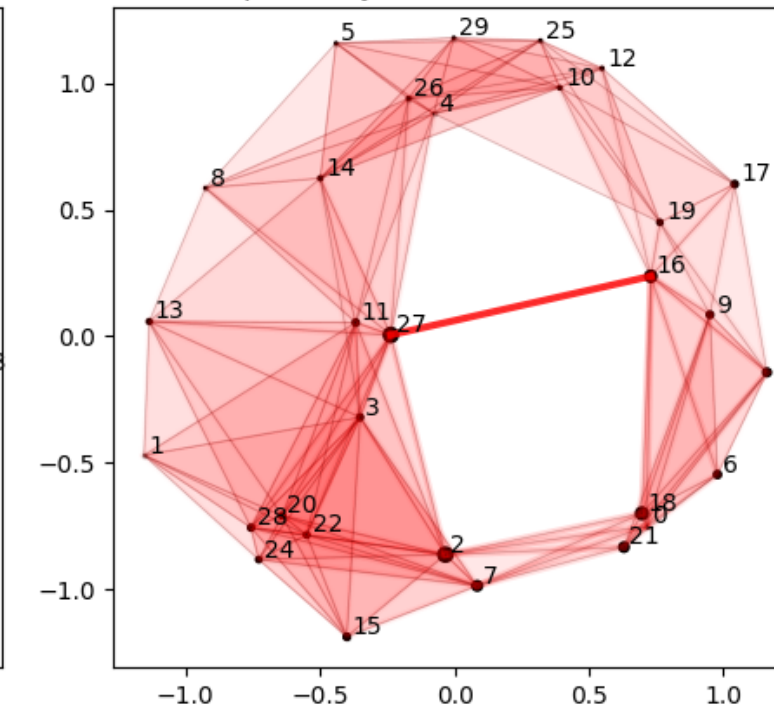
Optimal cycle at 224, [224, 227)



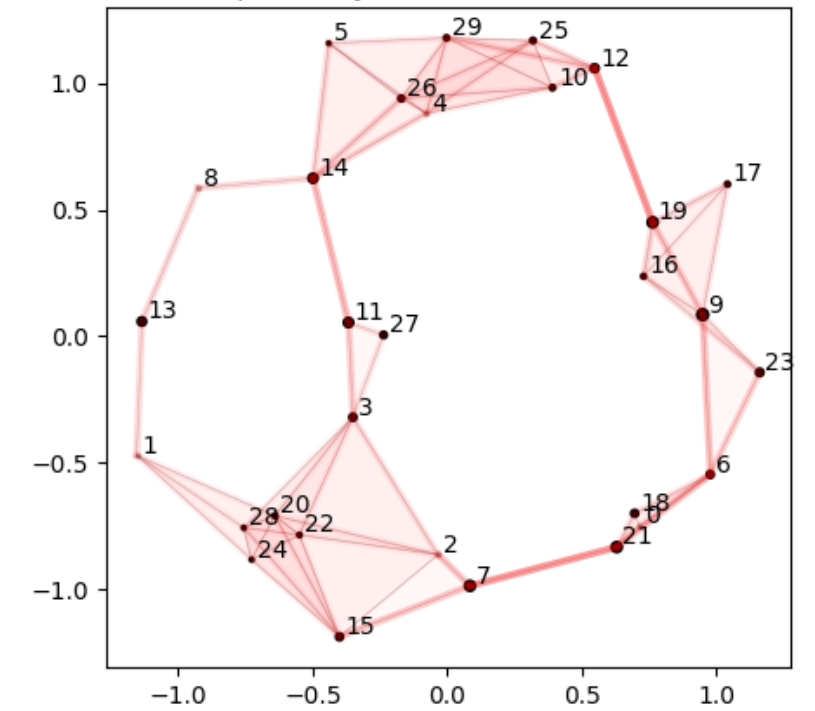
Optimal cycle at 132, [132, 299)



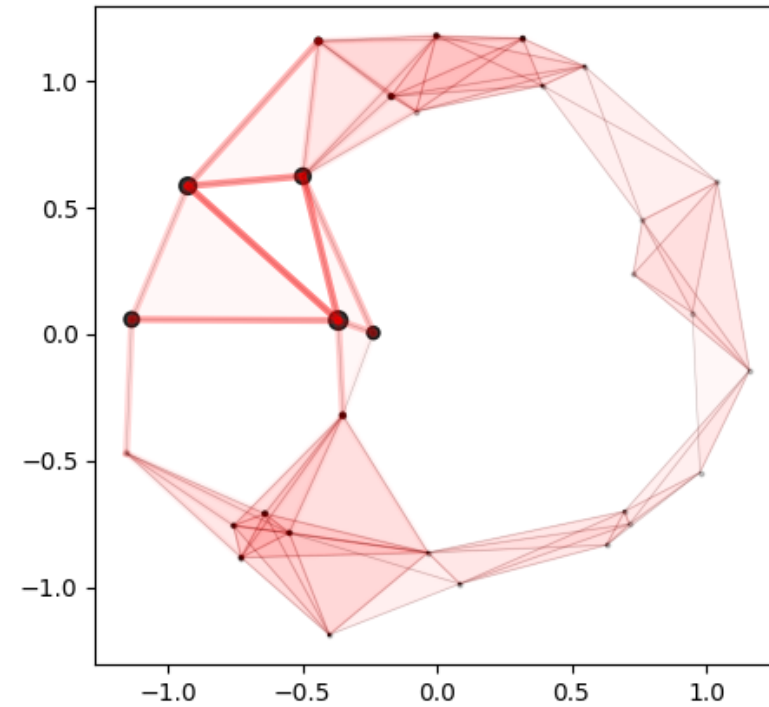
Optimal cycle at 413, [413, 467)



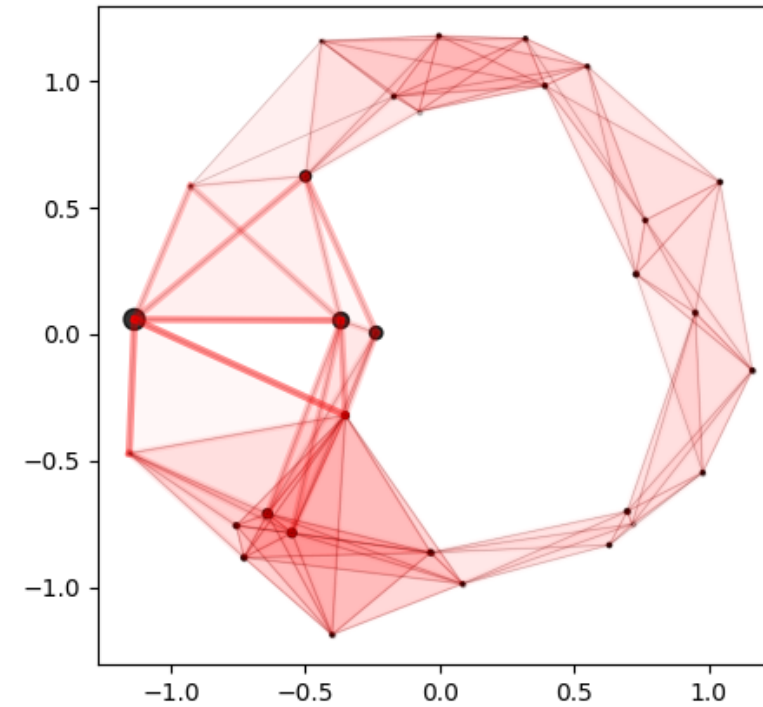
Optimal cycle at 147, [147, 612)



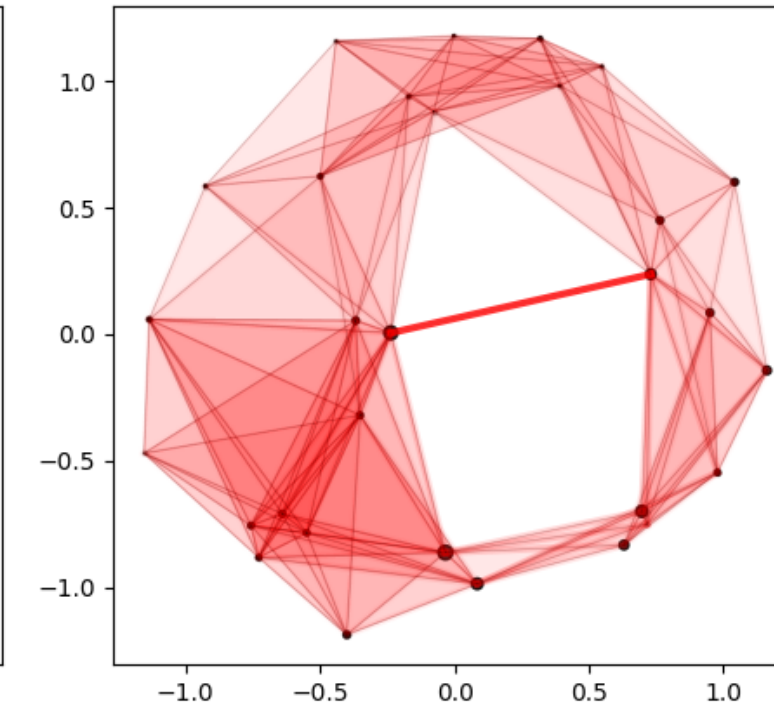
Optimal volume cycle at 226, [224, 227)



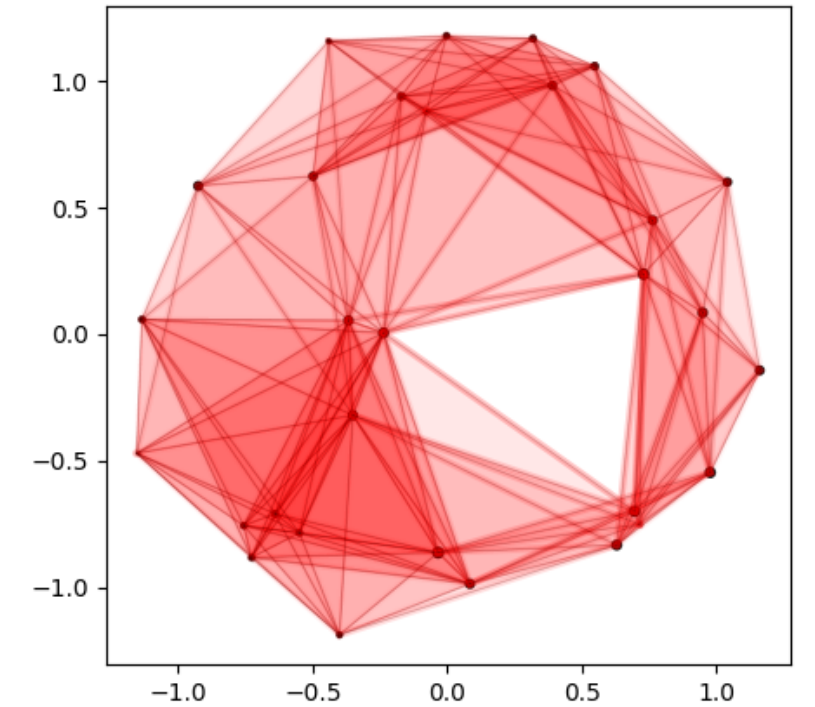
Optimal volume cycle at 298, [132, 299)



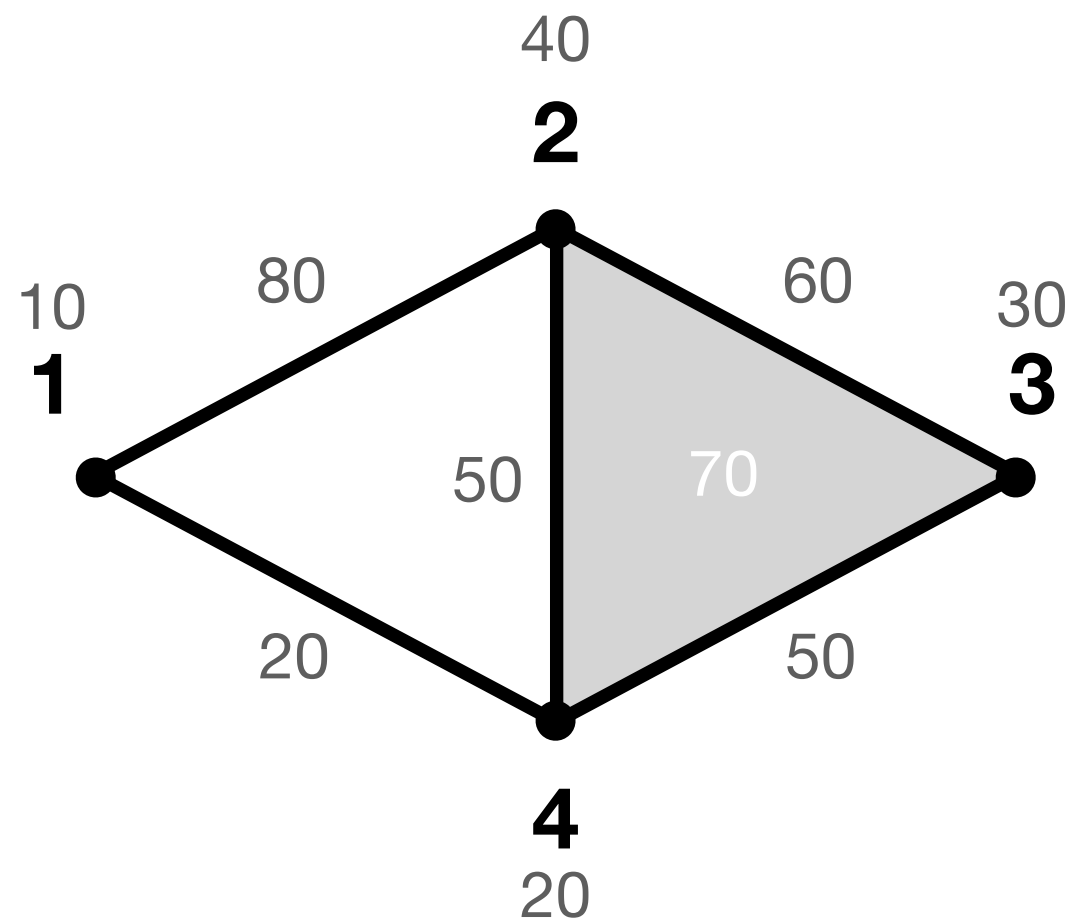
Optimal volume cycle at 466, [413, 467)



Optimal volume cycle at 611, [147, 612)



Boundary matrix at filtration step t



B =

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

Filtration function provides order on simplices, therefore on columns and rows of the filtration matrix. Ties are broken, first by simplex dimension, second by lexicographic order given by order on vertices.

Boundary matrix at filtration step t

10
1


	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

dt=1

Boundary matrix at filtration step t

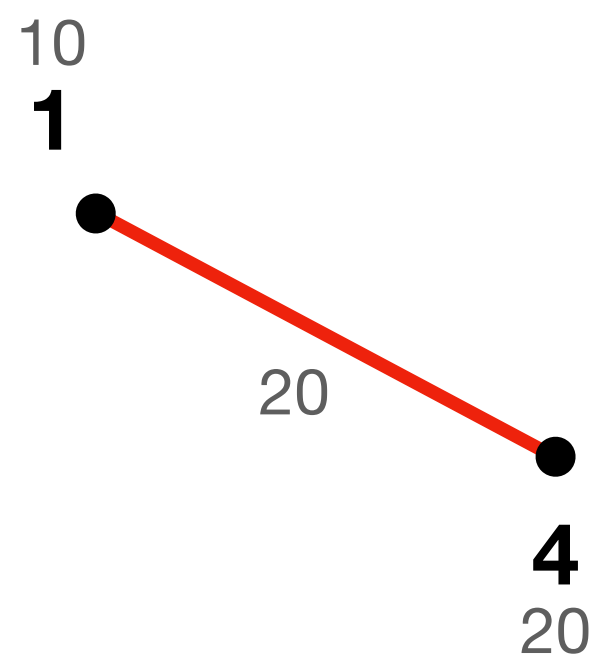
10
1
●

●
4
20

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

dt=4

Boundary matrix at filtration step t

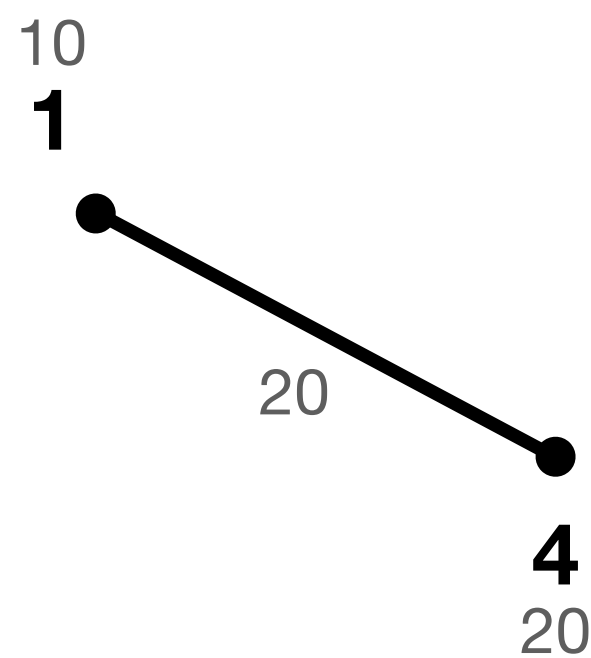


	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

dt=14

$B_1(t)$

Boundary matrix at filtration step t

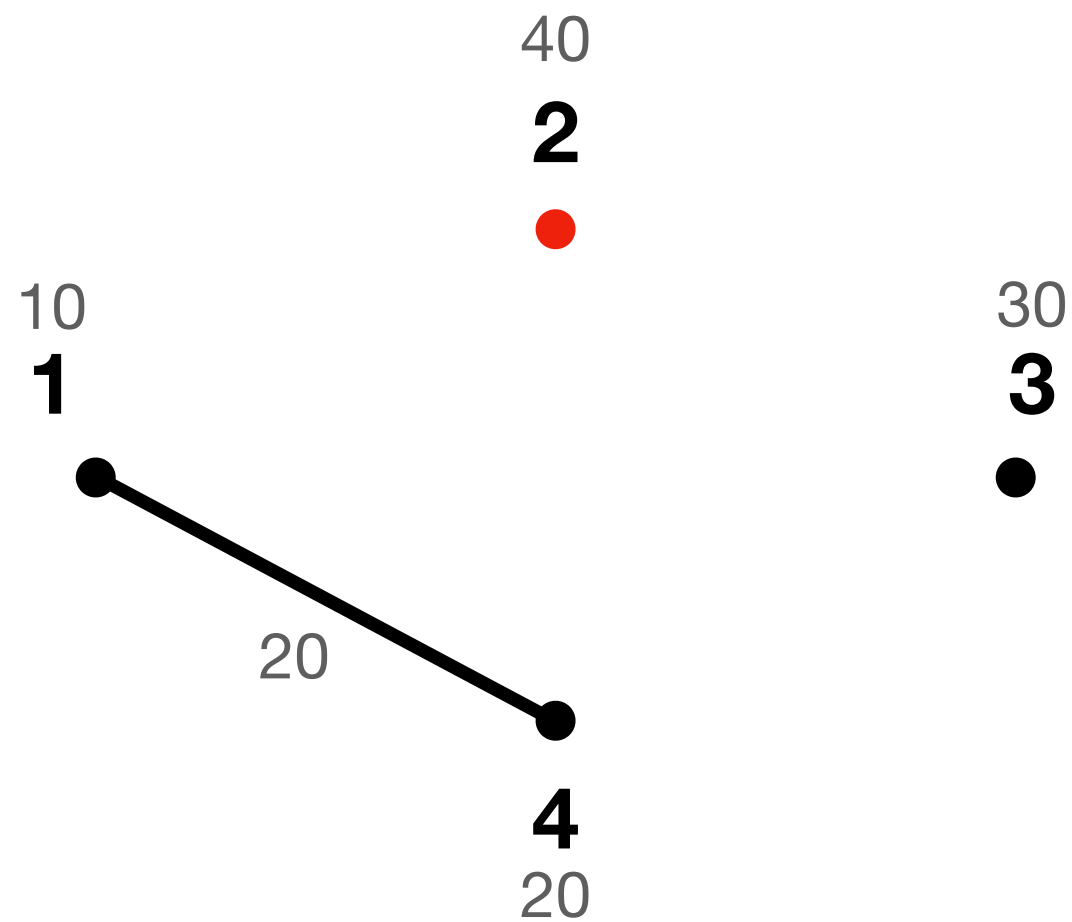


	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

dt=3

$\mathbf{B}_1(t)$

Boundary matrix at filtration step t

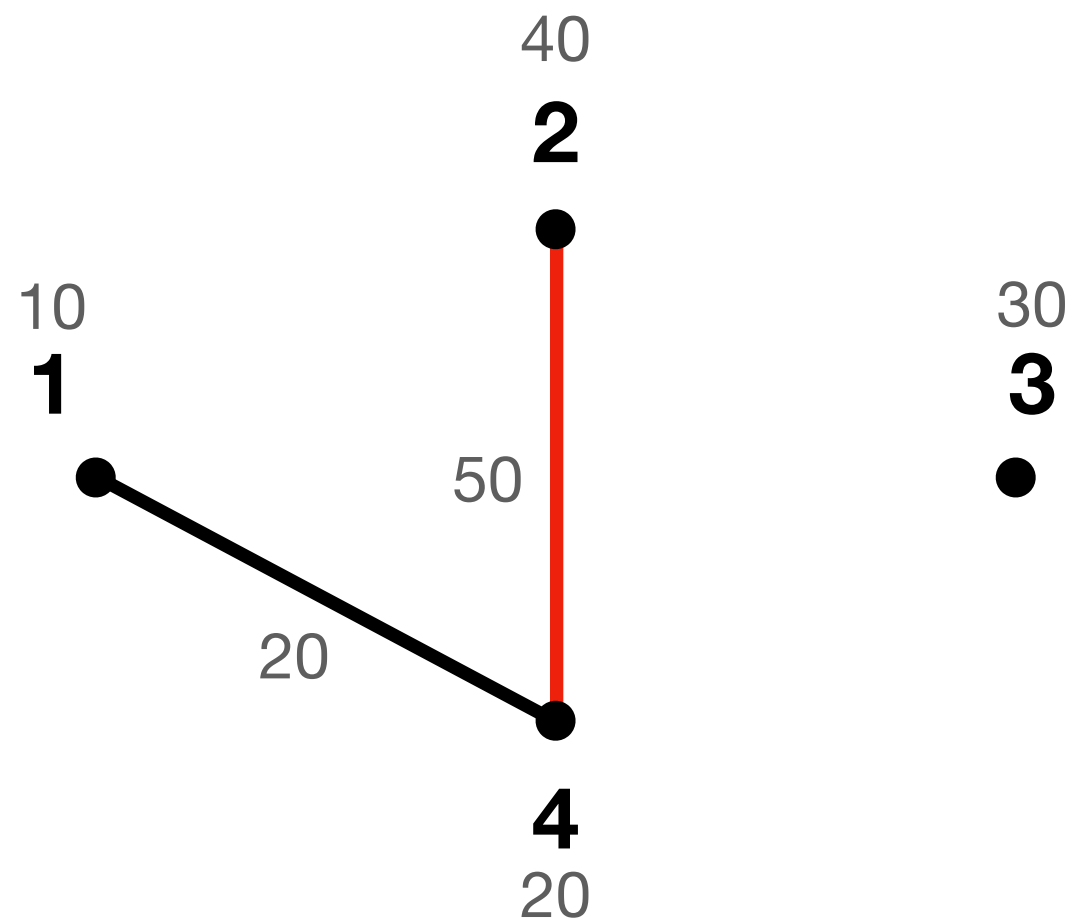


	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

dt=2

$B_1(t)$

Boundary matrix at filtration step t

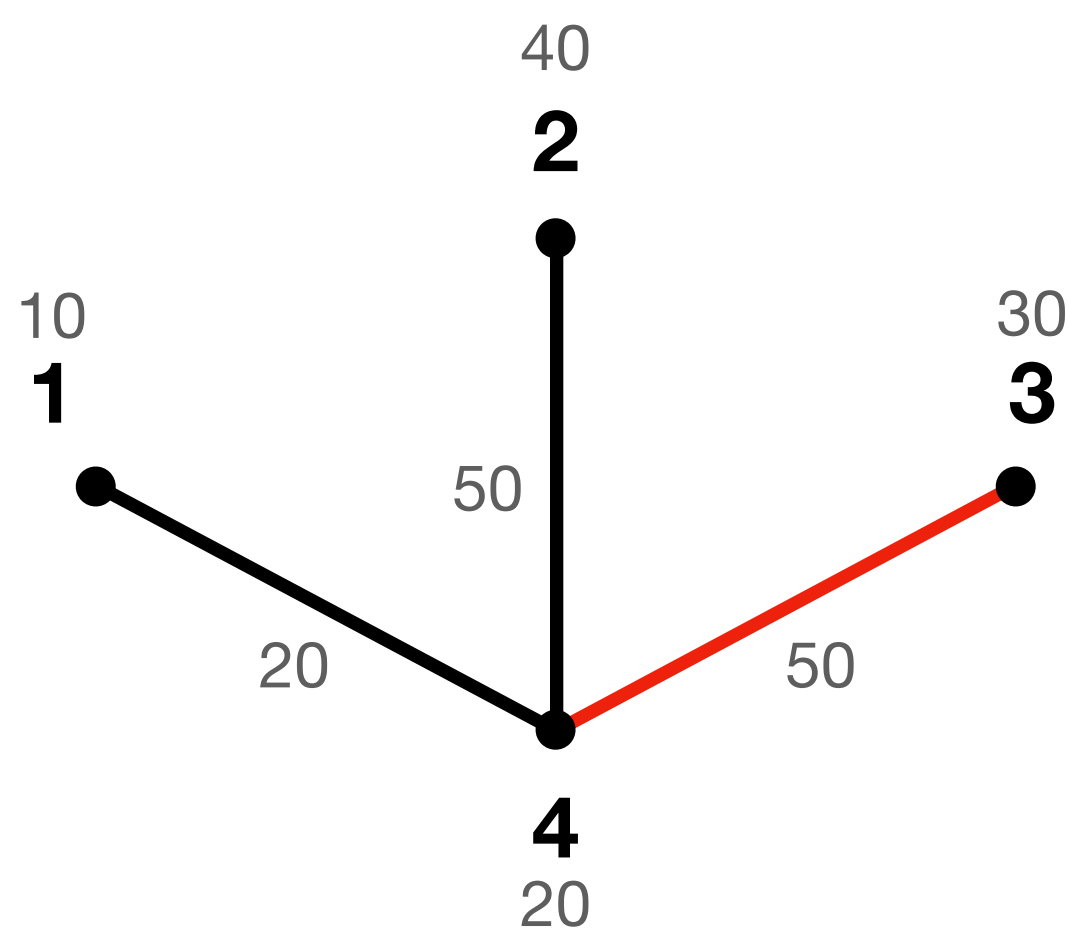


	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

dt=24

$B_1(t)$

Boundary matrix at filtration step t

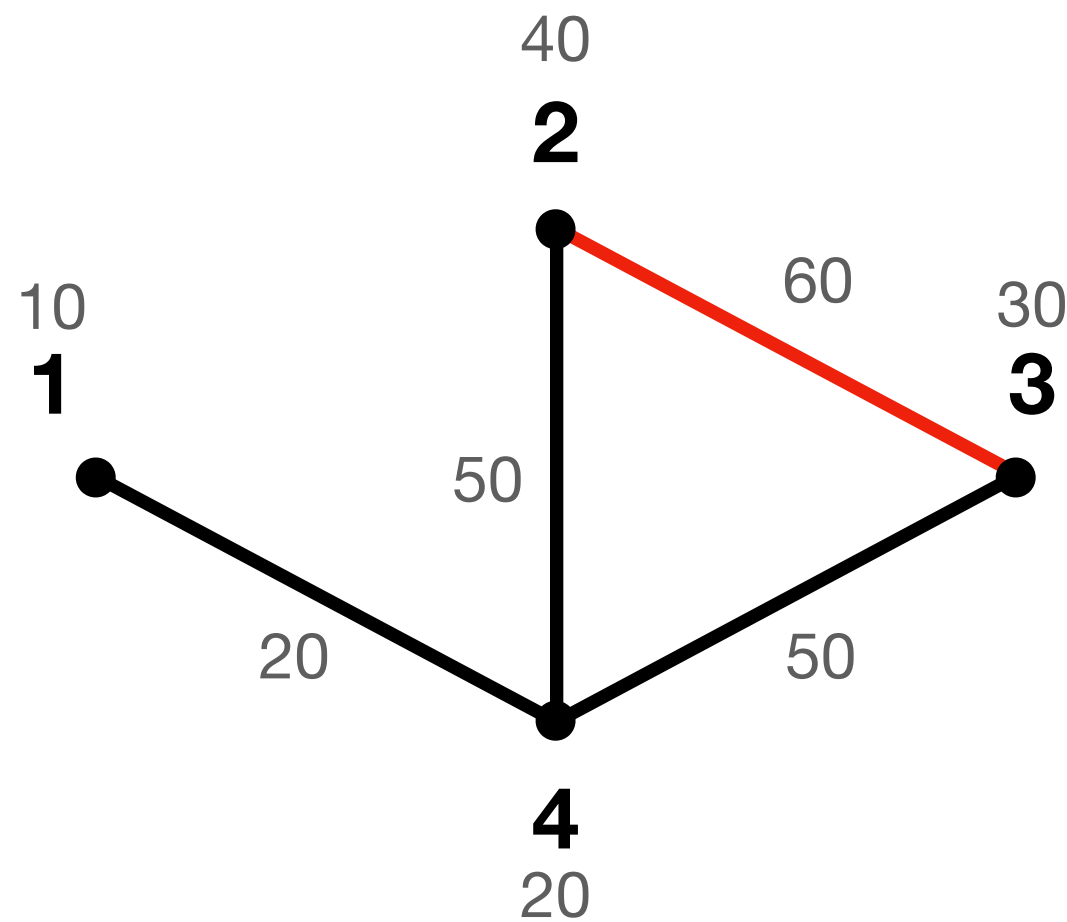


	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

dt=34

$B_1(t)$

Boundary matrix at filtration step t

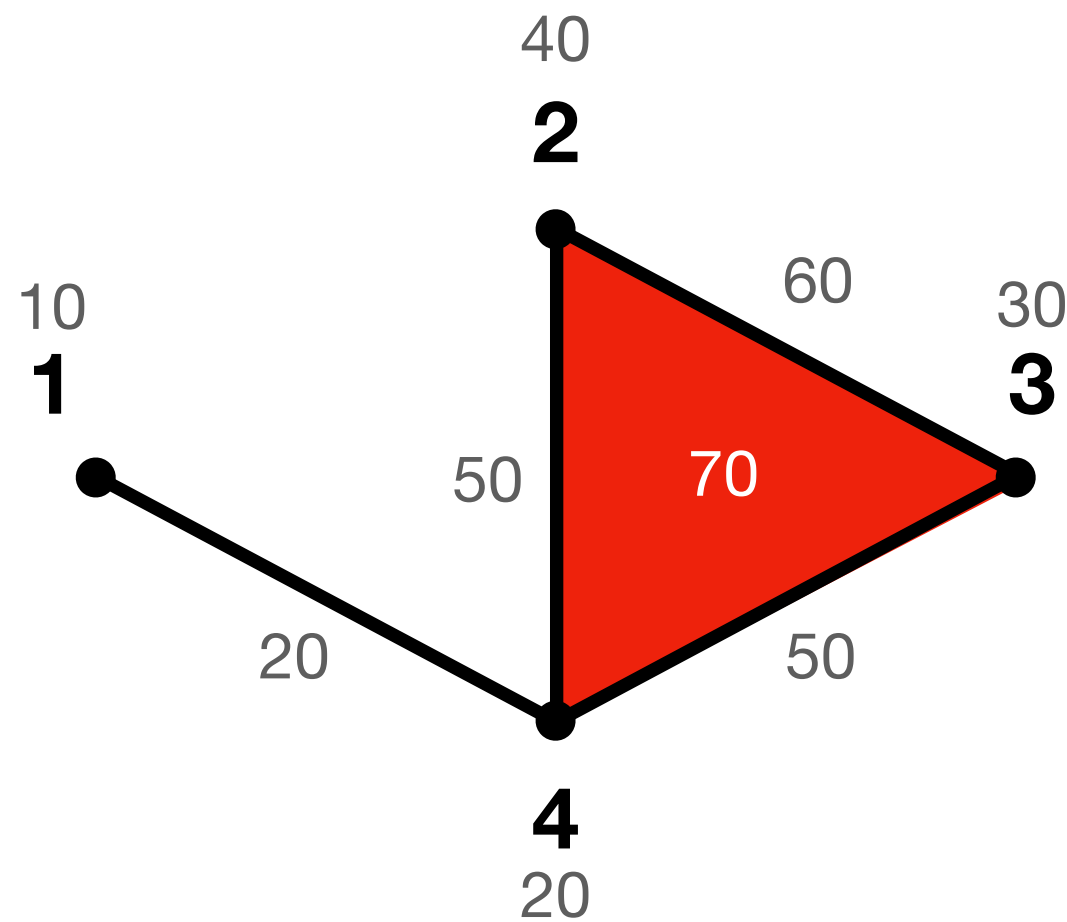


	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

dt=23

$$\mathbf{B}_1(t)$$

Boundary matrix at filtration step t



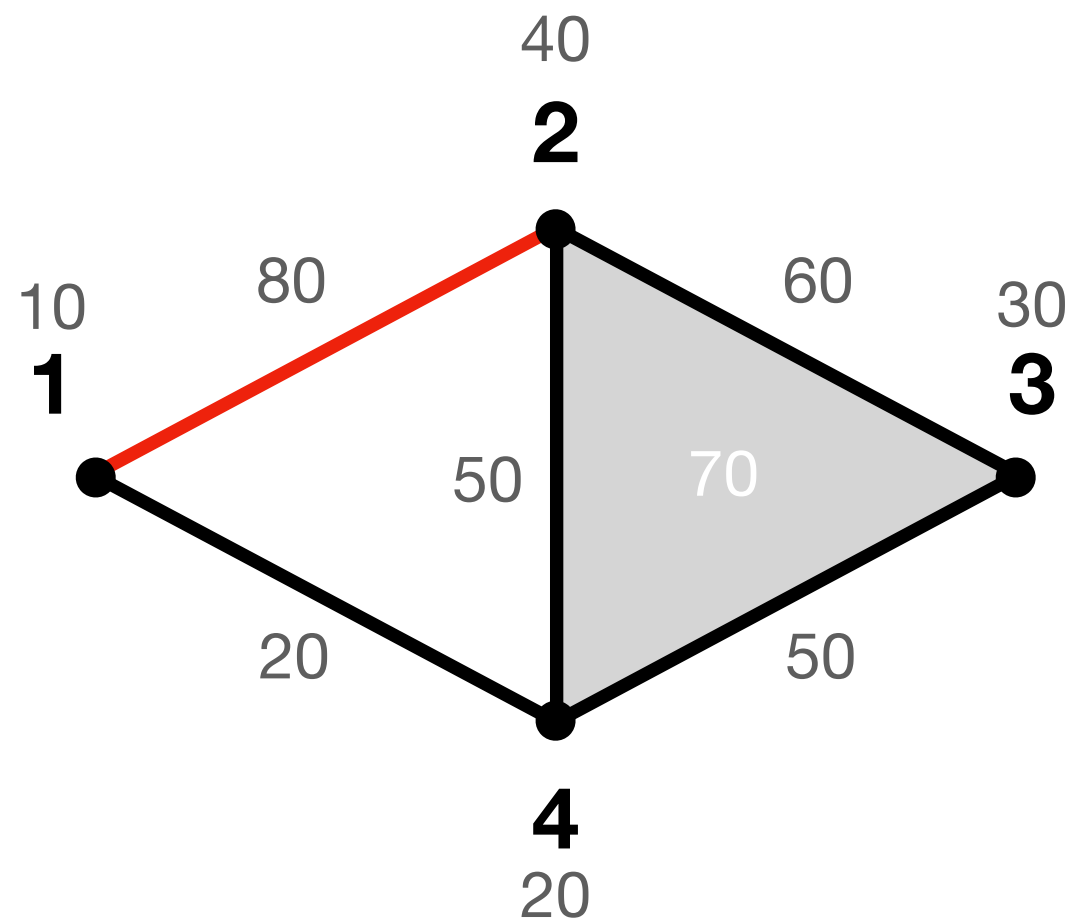
	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

dt=234

$B_1(t)$

$B_2(t)$

Boundary matrix at filtration step t



	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							1
4			1			1	1			
14										
3							1	1		
2						1		1		1
24									1	
34									1	
23									1	
12										

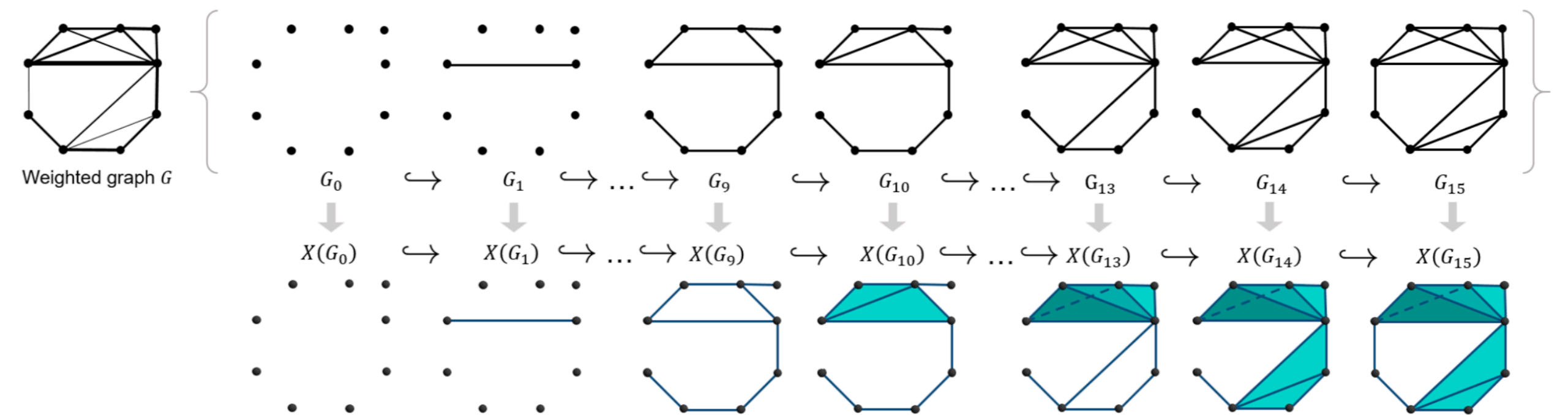
dt=12

$B_1(t)$

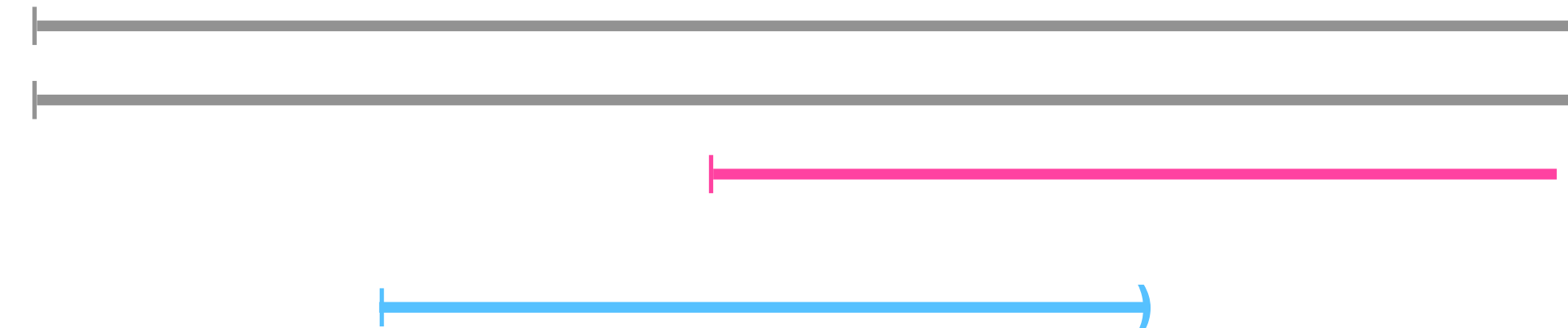
$B_2(t)$

Persistent harmonic representatives

Filtration

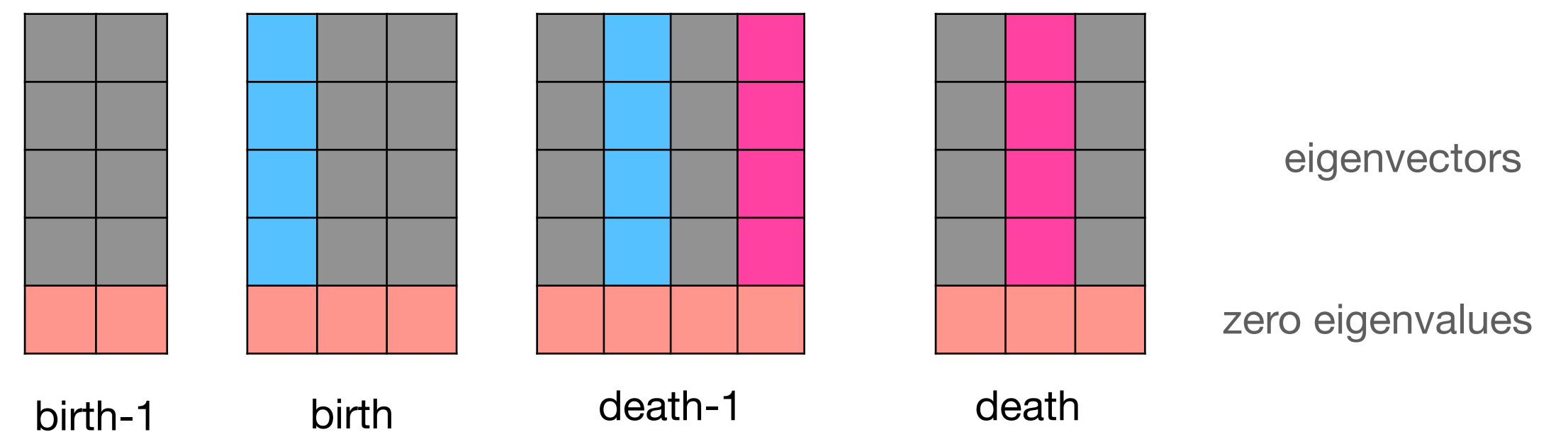


Persistence barcode



Higher-order Laplacian at filtration step t

$$\mathbf{L}_k(t) = \mathbf{B}_k(t)^T \mathbf{B}_k(t) + \mathbf{B}_{k+1}(t) \mathbf{B}_{k+1}(t)^T$$



Persistent harmonic representatives

Solve two eigenproblems on $\mathbf{L}_{k,t}$ with $t = b_i$ and $t = b_i - 1$ to find two sets of eigenvalues $\lambda_{k,t}$ and corresponding eigenvectors $\mathbf{V}_{k,t}$ of dimension k at time t to obtain zero eigenvalues (up to a tolerance hyperparameter δ) and their corresponding eigenvectors

- eigenvalues λ_{k,b_i-1}^0 and corresponding eigenvectors $\mathbf{V}_{k,b_i-1}^0$ at time $b_i - 1$ immediately preceding birth time b_i ,
- eigenvalues λ_{k,b_i}^0 and corresponding eigenvectors $\mathbf{V}_{k,b_i}^0$ at birth time b_i .

Find the representative vector $\mathbf{w}_{k,b_i}^+$ at birth such that $\mathbf{w}_{k,b_i}^+ \in \mathbf{V}_{k,b_i}^0$, but $\mathbf{w}_{k,b_i}^+ \notin \mathbf{V}_{k,b_i-1}^0$. To do so,

$$\mathbf{w}_{k,b_i}^+ = \arg \max_{\mathbf{v}_{k,b_i}^0 \in \mathbf{V}_{k,b_i}^0} \min \sigma (\mathbf{V}_{k,b_i-1}^0 \mid \mathbf{v}_{k,b_i}^0[: -1]) , \quad (11)$$

where $\min \sigma (\mathbf{X} \mid \mathbf{x})$ is the smallest singular value of the augmented matrix $\mathbf{X}$ by a vector $\mathbf{x}[: -1]$, i.e. with the last element truncated.

Higher-order Laplacian at filtration step t

$$\mathbf{L}_k(t) = \mathbf{B}_k(t)^T \mathbf{B}_k(t) + \mathbf{B}_{k+1}(t) \mathbf{B}_{k+1}(t)^T$$

